

The Winner's Curse in Test-Blind Admissions^{*}

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Should universities admit students without considering performance on their final exams? When universities do so, students who accept the offers may underperform for two reasons: students may reduce their academic effort toward the test (moral hazard), or students with low private signals may self-select into accepting offers (private information). We disentangle these forces in the context of the UK's university application system using the staggered rollout of unconditional offers, which allowed some students to matriculate at university without any requirements on their age-18 final exams (A-levels). Students who accept an unconditional offer underperform those who decline one by 0.14SD on age-18 exams, conditional on observables. Using a two-way fixed effects specification, we find that exposure to unconditional offers has precisely zero impact on students' age-18 exam performance, indicating that the observed gap is driven by private information, not moral hazard. Exposure to unconditional offers increases university matriculation by 1.1 percentage points. To determine the overall policy implications, we develop and estimate a structural model to evaluate the spillover effects onto non-recipients, the value students place on the certainty of admission, and the extent of misallocation. We estimate that students know two-thirds of the variance in their realized test scores before taking the test. The median student does not value the certainty of admission. Unconditional offers decrease total student welfare by 0.7% relative to a ban and redistribute welfare from the three highest-achieving quartiles to the lowest-achieving quartile. A shift to post-qualification admissions, under which students apply after receiving their age-18 exam scores, raises student welfare by 12.5% in our simulations. All students except the lowest-achieving quartile benefit, as do all programs except the most-selective quartile. Our results highlight how the distributional effects of admissions policies depend on student information, informing current policy debates in the UK.

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1 Introduction

In 2024, nearly three million students across the US, France, and Japan — 79% of these countries' college entrants — were admitted to college before or without a final exam score.¹ When universities choose which students to admit, it can change the course of those students' lives — and thousands of colleges do so without considering a key component of a student's academic record. These test-optional and test-blind systems often allow students to know if and where they can attend university much earlier, but may also lead to students studying less. Further, universities that use test-optional or test-blind admissions have fewer ways to evaluate students, making their selection problem more difficult. Media narratives and research on this topic have largely focused on the use of standardized tests at elite colleges, but these policies are relevant for students across the ability distribution and across the world.

Whether universities should or should not use final exams to admit students depends on the answers to three questions. When university entrance no longer depends on exams, do students learn less? How do test-blind admissions change if and where students attend university, considering both direct impacts on students subject to these policies and spillovers to other applicants? And finally, given what students value — including both match quality and certainty — are there admissions systems which outperform a test-blind mechanism?

We answer these questions with evidence from universities in the United Kingdom, which used a form of test-blind admissions during the 2010s. After applying to programs, students receive tentative offers which are conditional on age-18 standardized test scores (A-levels); they convert to admittance so long as students meet the relevant conditions. In 2014, some programs began making *unconditional offers*; rather than having admittance rely on future age-18 test scores, programs guaranteed admittance for students regardless of their scores on the exam. Students who ranked these offers gained certainty about their placement, but possibly at the expense of attending a more-selective program if they earned high grades. Policymakers criticized their rapid proliferation, concerned that these offers shifted students toward worse universities and reduced students' academic effort in studying for their exams. These offers were eventually banned by universities themselves in 2021.

Unconditional offers were used by academically less-selective programs to enroll higher-ability students, but the expected gains in student achievement did not materialize. These programs targeted unconditional offers toward high-achieving students with 0.4SD higher teacher-predicted grades on average. Extrapolating from students' high academic characteristics observed at the time of offer, programs would have expected the realized age-18 test scores of incoming students to increase by 0.06SD in the year they began using these offers. However, enrolling students, conditional on observables, scored 0.14SD *lower* on their age-18 exams than students who received the offer but declined. As a result, programs experienced a *decrease* in achievement of 0.02SD when they began making unconditional offers.

¹In the US, we count students attending community colleges or students enrolling at other colleges without submitting the ACT or SAT. In France, we count *lycée* students who receive *Parcoursup* offers before taking their *baccalauréat* exams; offers are contingent only on passing this exam. In Japan, we count students who enroll in private universities through the *sōgō-gata* or *gakkō suisen-gata* selection paths, which rarely require the Common Test for University Admissions.

This taker–decliner achievement gap is consistent with two factors. There is a *moral hazard* explanation: students may reduce test-taking effort after taking an unconditional offer, knowing that their end-of-year grades no longer matter. On the other hand, students may have *private information*: students who expect to do poorly on their end-of-year exams, and so to miss the grade requirements of their other offers, may self-select into taking these offers. Critics, including the Education Secretary, argued that these offers reduced students’ effort and distorted students’ decisions, and Universities UK cited lower motivation among takers when it restricted them in 2021. Effectively, these narratives claim that the taker–decliner gap was entirely due to moral hazard, even though it may have also been caused by private information.

We distinguish between these two explanations with reduced-form evidence. To test for moral hazard, we estimate the effect on age-18 test scores of unconditional offer *exposure*, defined as applying to a program that uses these offers. We focus on the first year that an application set is exposed to unconditional offers; first-year exposure is plausibly exogenous if students are unaware of when programs begin making these offers, and we provide evidence consistent with this assumption. We estimate the effect of first-year exposure on test scores using a two-way fixed effects design which controls for students’ application sets, interacted with an indicator for post-first-year usage to isolate first-year treatment effects.

Using this specification, we find that exposure to unconditional offers is inconsistent with the moral hazard hypothesis. We estimate a statistically precise null effect of unconditional offer exposure on later test scores. This finding is robust to a battery of alternative specifications. We conclude that contrary to popular belief, the gap in achievement between unconditional offer takers and decliners is driven by private information, not moral hazard.

Though they did not affect test scores, unconditional offers did increase university attendance and the quality of universities attended. Students exposed to unconditional offers are 1.1 percentage points more likely to matriculate into university. This increase is driven by attendance at middle-tercile universities. Among the set of students that attend university, exposure to unconditional offers marginally raises the quality of university attended.

We then turn to the general equilibrium effects of unconditional offers, and consider how they affect students who do *not* receive unconditional offers, and if there are policies which are better for welfare. To answer these questions, we estimate a structural model of university applications. Our model includes both private student information and an unconditional-offer premium: the value a student places on an offer being guaranteed, whether from risk aversion, lower stress, or settling her plans early. Given the absence of moral hazard in our reduced-form results, we can simplify the model by assuming that students’ effort is not reduced in response to unconditional offers. The model features a four-stage process mimicking the UK university application system in which students hold private information about their own eventual exam performance. We recover the unconditional-offer premium and the extent of private information: the model estimates that the unconditional-offer premium is small for the average student, and that students know two-thirds of the variance of their age-18 test scores before taking them.

We then turn to estimating welfare under various admissions designs: a full ban on unconditional offers, unconditional offers used at their 2018 levels, and post-qualification admissions, under which students apply to university only *after* receiving their age-18 grades. Post-qualification

admissions were debated in the UK in 2021 and are again under discussion as AI makes predicted grades harder to defend (Harvey-Brewin, 2026). In aggregate, unconditional offers at 2018 levels decrease student utility by 0.7%, benefiting the lowest-achieving quartile of students at the expense of the other three. However, post-qualification admissions lead to an increase in welfare of 12.5%, largely through improved allocation, with benefits accruing to higher-achieving students. Programs largely benefit from post-qualification admissions, as well; however, the most-selective quartile of programs is marginally hurt, which may lower the chance of policy reforms occurring.

Our counterfactual simulations and reduced-form analysis contribute to the growing literature on the optimal design of college admissions systems. The shift to test-optional university admissions in the wake of COVID-19 — and the shift back to test-required admissions a few years later — have led to a rapid proliferation of papers analyzing the efficacy of such policies and impacts on students and cohort composition, mostly at selective institutions (Sacerdote, Staiger and Tine, 2025; Avery, Shi and Magouirk, 2025). Dessein, Frankel and Kartik (2025) focuses on universities’ choice to be test-optional or test-required; in their model, inter-university competition leads to an adversely selected applicant pool. Doğan and Erdil (2025) discusses the benefits of post-qualification admissions from a theoretical lens. Borghesan (2025) uses a structural model to derive equilibria when universities stop requiring the SAT. Our contribution here is twofold. We provide reduced-form evidence on test-blind admissions from less-selective universities, allowing us to test the broader impacts of such policies, and do so in a setting that informs the global debate.

Our setting also allows us to study how admissions policies induce or deter student effort. A small literature shows that affirmative action can raise effort exerted by high-school students who benefit from such policies (Grau, 2018; Akhtari, Bau and Laliberté, 2024; Cotton, Hickman and Price, 2022). A recent paper from Krishna et al. (2026) shows that changing grade reporting on centralized tests can also induce effort. Existing evidence on how test-blind policies change student effort is correlational (Office for Students, 2019); we provide the first causal estimate, separating the effort response from selection.

The timing of admissions has been studied through a large theoretical literature; in matching markets where students are screened by multiple schools, enrollees are adversely selected. Most directly, Chade (2006) develops a model with imperfect signals in which being matched leads to an “acceptance curse”; this is further developed by Lee (2009) in the university admissions setting.² In these papers, inter-university competition leads to adverse selection; in our setting, selection occurs primarily through students’ private information about a bad future signal. In contrast to Lee (2009), who shows that the winner’s curse leads to efficiency-improving early admissions, we demonstrate empirically that selection on students’ private information acts *against* early admissions. Finally, this literature is largely theoretical; we provide reduced-form evidence resulting from quasi-experimental variation in offer timing.

We also contribute to a literature on preference estimation in the context of application systems. We build upon the work of Agarwal and Somaini (2018), Larroucau and Rios (2020), Bleemer (2024), and Kapor (2025), which model students’ application choice of portfolios or rank-order lists. While the literature abstracts away from the timing of offers, we incorporate offer types into

²See also Chade, Lewis and Smith (2014) and Kloosterman and Troyan (2020); the latter focuses on student inferences after being accepted.

the student-side choice problem directly. Further, we discuss how preferences under correlated admissions systems (such as those in [Ali and Shorrer, 2025](#)) can be estimated using the methodology of [Larroucau and Rios \(2020\)](#), which explicitly requires independent admissions.

Our decomposition of the taker–decliner achievement gap relates to a literature focused on insurance and consumer credit differentiating adverse selection and moral hazard. The positive correlation test of [Chiappori and Salanié \(2000\)](#) describes how correlation between insurance take-up and outcomes is a sign of private information, but distinguishing the two forces requires additional exogenous variation, such as the field experiment studied by [Karlan and Zinman \(2009\)](#).³ We bring this literature to the new domain of college admissions.

2 Empirical Setting, Data, and Descriptive Results

2.1 University applications in the UK

In many countries, including Germany, Australia, and China, universities use *post-qualification* admissions systems, meaning that students take local or national standardized examinations and only after receiving the scores do students apply to and enroll in universities.⁴ In this sense, both students and universities finalize the matching process after all the relevant information (i.e., test scores) are observed by both parties.

In contrast, the United Kingdom uses *pre-qualification* admissions. Students apply to university in January during their final year of secondary school (Year 13) with grades from exams taken at age 16, as well as teacher-predicted grades for age-18 exams that students have not completed when they apply. Students receive offers and rank choices before taking their exams and receiving their final grades. As a result, programs have historically made “conditional offers”, which are contingent on students achieving certain grades in their final exams; programs can withdraw offers to students who do not meet these conditions. This means that students face uncertainty about their university enrollment until they receive their final grades.

Approximately 250,000 18-year-old students from England, Wales, and Northern Ireland⁵ apply every year to university through the centralized Universities and Colleges Admissions Service (UCAS) platform. Because tuition fees for domestic students are uniform across universities⁶, the relevant tradeoffs for UK students are over academic fit and preferences for geography, amenities, and other factors. [Table 1](#) summarizes the process as of 2012.⁷

³See also the canonical models of [Akerlof \(1970\)](#) and [Rothschild and Stiglitz \(1976\)](#); further theoretical work in [Chiappori et al. \(2006\)](#); and [Adams, Einav and Levin \(2009\)](#), [Einav, Finkelstein and Cullen \(2010\)](#), and [Einav and Finkelstein \(2011\)](#) for additional empirical studies.

⁴In Germany, this is the *Abitur*; in Australia, the *ATAR*; in China, the *gāokǎo*.

⁵We exclude Scottish students, as they face different exam and tuition fee systems to students in the rest of the UK.

⁶Tuition fees for domestic students are capped in England, Wales, and Northern Ireland, and essentially all universities charge fees at the cap. In England and Wales, the cap was set at £9,000 per year in 2012, and following irregular below-inflation increases, it currently stands at £9,790. Students have access to standardized government loans with income-contingent repayment plans for tuition and living costs.

⁷There have been several changes in the university applications process during our analysis window. In 2007, the first year of our data, students could apply to up to 6 programs. This was reduced to 5 in 2008. In the late 2010s, the UK government phased out an intermediate exam taken at age 17 prior to university applications; refer to [Tadjfar and Vira \(2025\)](#) for more details.

Table 1: Timeline of UK university applications process as of 2012

January	Students apply to up to 5 programs, specific majors at specific universities. ^a Applications include age-16 and age-17 standardized test grades and teacher predictions of age-18 grades.
May	Programs make offers of places. Offers are contingent on the student meeting conditions on their age-18 grades.
June	Students rank up to two programs from which they receive offers; all others are forfeited. ^b Students then take age-18 exams.
August	Age-18 exam results are released, and students match to programs. Programs must accept all students who ranked their offer and met the offer conditions, and have discretion to accept students who failed to meet conditions. Students who are not accepted at either ranked program may go unmatched or join an after-market clearing process. ^c

^a There are approximately 35,000 such programs across 160 different universities.

^b These are termed the *firm* and *insurance* choice; we call them the first and second choice throughout.

^c 71% of students who ranked programs are admitted to their first choice, 10% to their second choice, 10% go unmatched, and 9% are admitted to a different program through the clearing process.

2.2 Unconditional offers

In the 2010s, universities increasingly began using *unconditional offers*, a form of test-blind admissions. In contrast to conditional offers, an unconditional offer admits students to a program regardless of their age-18 exam results. Historically, these offers were primarily reserved for students applying to university after completing secondary schooling (such as students re-applying after a gap year), as these students have already completed their exams and their final grades are included in the initial application. In 2014, programs increasingly began to make unconditional offers to applicants who had *not* completed their exams.⁸ Programs also began to make *binding unconditional offers*,⁹ where the offer had no conditions only if the student ranked it as their first choice. We pool together these two types of unconditional offers and refer to them as unconditional offers throughout. Figure 1 shows the rapid growth in unconditional offers from 2010 through 2019 for our analysis sample: in that final year, 38% of applicants received at least one unconditional offer.

Reactions to the growth of unconditional offers varied greatly. Critics, including the Education Secretary and the Association of School and College Leaders,¹⁰ argued that offers were primarily a tool for universities to increase enrollment rather than a genuine benefit to students, and that the pressure to commit to a university before sitting age-18 exams discouraged academic effort.¹¹ A particular concern was that binding unconditional offers pressured students to forfeit their higher-

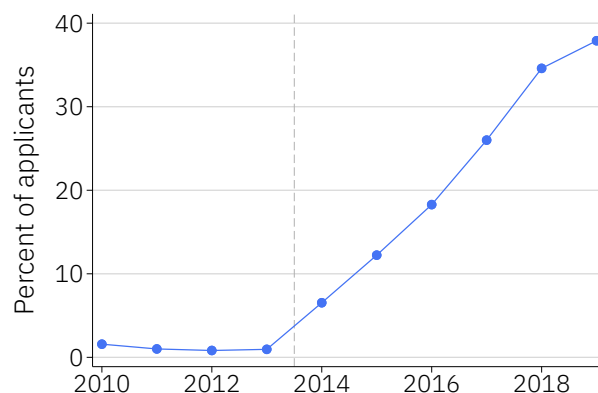
⁸One reason behind this change is the lifting of strict enrollment caps. In 2014, enrollment caps were lifted at all universities (except alternative providers: private “for-profit” entities or other independent institutions that receive no UK government funding; they enroll less than 2% of UK undergraduates and are not considered in our results). In 2015, enrollment caps were lifted at all universities including alternative providers.

⁹In the UK, these were referred to as *conditional unconditional offers*, as they were conditional on the student ranking the offer first; we use the term ‘binding unconditional’ for clarity.

¹⁰In the UK, ‘college’ refers to secondary education institutions such as sixth form colleges and vocational colleges, equivalent to grades 11 and 12 in the US; they do *not* refer to universities or other tertiary education institutions.

¹¹See, for example, coverage in *The Guardian* in 2018 and 2019.

Figure 1: Percent of applicants receiving an unconditional offer, 2010–2019.



quality options, effectively limiting their ability to benefit from strong age-18 performance. Universities defending the practice argued that unconditional offers helped them attract high-achieving students and offered genuine reassurance to anxious applicants; a 2019 UCAS survey found that 70% of applicants supported unconditional offers, highlighting a reduction in stress levels from the certainty of placement.

The widespread usage of unconditional offers ended in 2021 amidst various bans. In 2020, amid concerns about pressure on students and disruption caused by the COVID-19 pandemic (which led to the cancellation of age-18 examinations that year), the UK government temporarily banned binding unconditional offers. Following this, in 2021, Universities UK, a group representing 142 universities, introduced a “Fair Admissions Code of Practice.” Under this code, universities could no longer make unconditional offers to students who had not yet completed their age-18 examinations, effectively ending the practice for mainstream applicants.¹² Universities UK cited several reasons for this self-imposed ban, most notably the “unfair pressure” created by binding unconditional offers and that unconditional offer takers were “less likely to report feeling motivated to perform well at school” (Universities UK, 2020).

2.3 Data sources

Our analysis draws on the Longitudinal Education Outcomes (LEO) database, produced by the UK Department for Education (DfE) and accessed via the Secure Research Service (SRS) of the Office for National Statistics, which allows us to link three administrative datasets from different data providers using anonymized individual identifiers (Office for National Statistics, 2025, 2026a,b).¹³

We observe the universe of university applications to UK universities between 2007 and 2021 in the Universities and Colleges Admissions Service (UCAS) dataset. The application data includes the program identifiers (including both university and major), whether each program made an offer and the type of offer (conditional or unconditional), the student’s ranking of their offers (first and second choices), and whether the student was ultimately admitted. The data also includes

¹²The [code of practice](#) also introduced requirements around transparency in admissions criteria and restrictions on the use of other incentive-based admissions practices.

¹³Due to the sensitive nature of this data, all analysis was conducted in person in the UK at the Institute for Fiscal Studies or at SRS-designated Saferooms or Safepods.

teacher-predicted and realized age-18 grades (A-levels, BTEC examinations), and the subjects students take in Years 12 and 13. Programs rely heavily on teacher-predicted age-18 grades to make initial offers, but recognize that these predictions are upward-biased and less predictive of university performance than realized scores (Murphy and Wyness, 2020). The National Pupil Database (NPD) provides student-level data on educational attainment for students in England completing secondary school between 2009 and 2021. The NPD data provides additional demographic characteristics, age-16 standardized test results (GCSEs), age-17 results (AS-levels), and realized age-18 grades (A-levels, BTEC examinations). Our research also uses data from the Higher Education Statistics Agency (HESA), which provides student-level data on university enrollment, degree program, degree type, program start and completion dates, and degree classification. The typical undergraduate degree lasts three years, after which students receive a degree classification — a coarse grade analogous to GPA, ranging from first-class honors through 2:1, 2:2, and third-class honors — which plays a significant role in graduate labor market outcomes (Britton et al., 2022). Our HESA sample spans academic years 2006–07 through 2020–21, and we observe outcomes for students matriculating from schools in England, Wales, Northern Ireland, and Scotland.

We restrict our main analysis sample to first-time applicants in the standard UCAS main scheme aged 18 as of August 31 of the application year. We also exclude students who apply only through alternative UCAS routes, or who cannot be linked to HESA data.¹⁴ We limit our analysis to students applying in the 2010–2019 application years. For the reduced-form analysis in Section 3, we further restrict our sample to students with application sets in which no constituent program had used unconditional offers in or before 2013 and all constituent programs each averaging at least 20 offers per year between 2010 and 2012. Additional details on the construction of variables are provided in Appendix B.

2.4 Targeting and take-up

The rise in unconditional offers is driven by smaller and academically less-selective programs. Table A.1 reports characteristics of programs where more than 5% of offers made were unconditional in some year between 2014 and 2019. Programs making unconditional offers were 39 percentage points *less likely* to be in the top tercile of university rankings, 11 percentage points less likely to be STEM programs, admitted students in 2010–2012 whose age-18 exams scored 0.5SD lower, and had lower historical rates of being ranked as a student's first choice.

What types of students did programs target with unconditional offers? Columns (1)–(3) of Table 2 show the mean age-16 and age-18 grades (converted to z-scores) of students who received unconditional offers, compared to those who did not. Students who received unconditional offers were stronger than non-recipients on all academic characteristics; the largest difference occurs in teacher-predicted age-18 scores, where recipients scored 0.40SD higher than non-recipients. However, the difference between takers and decliners among recipients indicates that on the basis of observables, weaker students self-selected into taking unconditional offers.¹⁵ This gap is

¹⁴UCAS programs that do not link to HESA data are generally specialist programs at arts academies or music conservatoires rather than traditional university programs.

¹⁵We define take-up as ranking a binding unconditional offer first, or ranking a standard unconditional offer first or second.

especially large in the realized age-18 exam scores. As reported in column (6), unconditional offer takers scored 0.28SD lower on their age-18 exams than recipients who declined them; we call this the taker–decliner achievement gap.¹⁶

Table 2: Mean standardized test scores of unconditional offer recipients and takers

	Receiving an unconditional offer			Take-up, among offered		
	(1) No offer	(2) Received offer	(3) Difference	(4) Decliner	(5) Taker	(6) Difference
Age-16 English (z)	−0.007 (0.003)	0.086 (0.006)	0.093*** (0.007)	0.129 (0.013)	0.076 (0.006)	−0.053*** (0.015)
Age-16 math (z)	0.022 (0.003)	0.062 (0.007)	0.040*** (0.008)	0.114 (0.015)	0.050 (0.007)	−0.064*** (0.017)
Teacher-predicted age-18 (z)	−0.299 (0.003)	0.101 (0.006)	0.400*** (0.007)	0.251 (0.014)	0.068 (0.007)	−0.183*** (0.015)
Achieved age-18 (z)	−0.282 (0.003)	−0.037 (0.007)	0.245*** (0.008)	0.191 (0.014)	−0.089 (0.007)	−0.280*** (0.016)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Means, with standard errors in parentheses, among students in the first year in which any program in their application set makes unconditional offers. Columns (1)–(3) compare students who receive an unconditional offer with those who do not; columns (4)–(6) compare, among recipients, those who rank an unconditional offer against those who do not. A student ranks an unconditional offer if they rank an unconditional offer first or second, or a binding unconditional first. N is the full regression sample, covering all cohorts and exposure groups, and ranges from 1.33m to 1.63m across rows with the availability of each score.

2.5 Enrolling student composition

Programs targeted unconditional offers towards high-achieving students, but academically weaker students accepted them; we examine how these two forces affected programs’ actual enrolling class.

Our analysis focuses on how enrolling class composition changed between the first year of usage and the years before unconditional offer use. Define $Z_{st} \in \{0, 1\}$ as an indicator for whether program s gave more than 5% unconditional offers in year t , and let t_s^{first} be the first year in which $Z_{st} = 1$. Table 3 reports coefficients from estimating

$$Y_{st} = \beta_1 Z_{st}^{\text{first}} + \beta_2 Z_{st}^{\text{later}} + \alpha_s + \lambda_t + \varepsilon_{st}, \quad (1)$$

where Y_{st} is an outcome of program s in year t , and α_s and λ_t are program and year fixed effects. The exposure indicator is split in two: $Z_{st}^{\text{first}} = \mathbf{1}\{t = t_s^{\text{first}}\}$, the first year a program makes unconditional offers, and $Z_{st}^{\text{later}} = Z_{st} \mathbf{1}\{t > t_s^{\text{first}}\}$, every subsequent year in which it makes them. The sample is program-years between 2010 and 2019, excluding programs that made unconditional offers before 2014 or averaged fewer than 20 offers per year in 2010–2012. The coefficient β_1 effectively compares enrolling class composition in the first year of unconditional offer usage to the pre-unconditional-offer period.

Panel A of Table 3 suggests that programs started making unconditional offers in a “bad year” to offset fewer applications. In the year unconditional offers were first used, programs had 21.4 fewer

¹⁶ The same qualitative pattern holds, with a smaller magnitude, among students exposed in a later year; see Table A.2 in the appendix.

Table 3: Change in program characteristics after adopting unconditional offers

	(1) First year of unconditional offers	(2) Later years of unconditional offers
<i>Panel A. Applications, offers, and enrollment</i>		
Applications	-21.40*** (2.049)	-19.12*** (1.906)
Offers made	-19.87*** (1.736)	-15.94*** (1.614)
Enrollment	1.863*** (0.355)	3.856*** (0.330)
Rate ranked first	0.035*** (0.002)	0.039*** (0.002)
Offer rate	0.003 (0.002)	0.001 (0.002)
<i>Panel B. Test scores of the enrolling class (z)</i>		
Age-16 English	0.012** (0.005)	0.008 (0.005)
Age-16 math	0.012** (0.006)	0.010* (0.005)
Teacher-predicted age-18	0.070*** (0.006)	0.070*** (0.006)
Expected age-18	0.055*** (0.005)	0.056*** (0.004)
Achieved age-18	-0.024*** (0.006)	-0.042*** (0.006)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each row is a separate regression of the named program-year outcome on indicators for the first year and for later years in which the program makes unconditional offers, with program and year fixed effects; columns (1) and (2) report the two coefficients. Expected age-18 attainment is a forecast of the enrolling class's achieved score from the characteristics observed at the time of application, contrasting with actual age-18 attainment. N ranges from 45,183 to 46,826 across rows.

applications compared to years without unconditional offer usage. Despite receiving fewer applications, these programs saw a statistically significant *increase* in enrollment of 1.9 students while holding the aggregate offer rate (including both conditional and unconditional offers) constant.

Panel B shows that programs enrolled students with better academic attributes observable at the time of application. The first three rows of Panel B show that in their first year of making unconditional offers, programs enrolled students with 0.01SD higher age-16 English and math scores and 0.07SD higher teacher-predicted grades. Absent any student private information or moral hazard effects, projecting these observables onto realized age-18 test scores leads to expected age-18 scores that are 0.06SD higher, as shown in the fourth row of Panel B.¹⁷

However, the actual academic performance of these students did not match the expected improvement. The last row of Panel B shows that the realized age-18 test scores of enrolling students were in fact 0.02SD *lower* than in cohorts where programs did not use unconditional offers. This large discrepancy between *expected* age-18 grades and *realized* age-18 grades suggests that students either had substantial private information about how they will fare in the actual exams relative to their teacher-predicted grades and sorted into accepting the unconditional offers accordingly, exerted *less* effort preparing for the exams than they otherwise would have in the absence of an

¹⁷We estimate this by regressing actual age-18 grades on teacher-predicted age-18 grades, age-16 grades, and demographic variables using results from in-sample students between 2010 and 2013. The fitted values from this regression are used as the expected age-18 achievement.

unconditional offer, or a combination of both.

3 Causal Effects of Unconditional Offers

Table 3 poses a puzzle: in the year a program began making unconditional offers, its enrolling class was *stronger* on test scores observable at the time of offer-making than in other years, yet its realized age-18 scores were *lower*. The taker–decliner achievement gap in Table 2 shows this more directly: takers scored 0.28SD worse than decliners on their age-18 exams.

There are two reasons that the taker–decliner achievement gap alone is not enough to conclude that unconditional offers hurt students. First, this gap is descriptive, not causal, and may reflect selection. Second, it does not account for how unconditional offers change where students enroll. This section addresses both issues.

3.1 Effects on test scores

Three components make up the taker–decliner achievement gap: the causal change in achievement when a student takes an unconditional offer, holding observables and unobservables constant (moral hazard, equal to the average treatment effect on treated students); the difference in students’ unobservable characteristics (private information); and the difference in students’ observable characteristics (the composition effect). Formally, let T_i indicate take-up by student i , $Y_i(1)$ and $Y_i(0)$ be i ’s age-18 score with and without take-up, and X_i be the characteristics programs observe when making offers. The taker–decliner achievement gap δ decomposes as ¹⁸

$$\delta \equiv \mathbb{E}_Y[Y \mid T = 1] - \mathbb{E}_Y[Y \mid T = 0] \quad (2)$$

$$= \text{Moral Hazard} + \text{Private Information} + \text{Composition Effect}, \quad (3)$$

where

$$\text{Moral Hazard} \equiv \mathbb{E}_Y[Y(1) - Y(0) \mid T = 1],$$

$$\text{Private Information} \equiv \mathbb{E}_X[\mathbb{E}_Y[Y(0) \mid T = 1, X] - \mathbb{E}_Y[Y(0) \mid T = 0, X] \mid T = 1],$$

$$\text{Composition Effect} \equiv \mathbb{E}_X[\mathbb{E}_Y[Y(0) \mid T = 0, X] \mid T = 1] - \mathbb{E}_Y[Y(0) \mid T = 0].$$

Subscripts give the variable being averaged: $\mathbb{E}_Y$ averages age-18 scores and $\mathbb{E}_X$ averages characteristics, each among the students in its conditioning set, so the outer $\mathbb{E}_X[\cdot \mid T = 1]$ weights by takers’ characteristics.

Of key interest is the *conditional* taker–decliner achievement gap,

$$\delta_X \equiv \delta - \text{Composition Effect} = \text{Moral Hazard} + \text{Private Information}, \quad (4)$$

which represents the portion of δ that cannot be predicted solely on the basis of observable characteristics; in effect, δ_X is the taker–decliner achievement gap that came as a surprise to programs.

¹⁸ $\mathbb{E}_X[\mathbb{E}_Y[Y(0) \mid T = 1, X] \mid T = 1] = \mathbb{E}_Y[Y(0) \mid T = 1]$. The right-hand sum then simplifies to $\mathbb{E}_Y[Y(1) \mid T = 1] - \mathbb{E}_Y[Y(0) \mid T = 0]$, which equals δ since $Y = TY(1) + (1 - T)Y(0)$.

We estimate δ_x by estimating the gap in age-18 test scores between takers and decliners with increasingly strict controls for observable characteristics, and report results in Table 4. Conditioning on all characteristics observed at the time of application, the gap between takers and decliners shrinks to -0.139SD . If we “over-condition” by including controls for application sets (unobserved by programs), the gap shrinks to -0.127SD , indistinguishable from the -0.139SD gap with only observed characteristics. This correlation between take-up and outcomes conditional on observables indicates the presence of either private information or moral hazard; in Appendix C, we show that they cannot be disentangled without further information.¹⁹ As described in Section 2.2, programs and policymakers attributed the taker–decliner achievement gap to moral hazard.

Table 4: The taker–decliner achievement gap

	$\hat{\delta}$	$\hat{\delta}_x$		
	(1)	(2)	(3)	(4)
	No controls	+ Academic attainment	+ Demographics, school	+ Application-set FE
Uncond. take-up	-0.280^{***} (0.016)	-0.133^{***} (0.011)	-0.139^{***} (0.011)	-0.127^{***} (0.038)
Composition effect		-0.147	-0.141	-0.153
N	1,630,305	1,522,513	1,519,370	308,671

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column regresses the achieved age-18 score on an indicator for take-up — an unconditional offer ranked first or second, or a binding unconditional ranked first — adding, cumulatively, blocks of the characteristics a program observes when it makes the offer. Column (1) estimates the raw taker–decliner achievement gap $\hat{\delta}$; columns (2)–(4) estimate $\hat{\delta}_x$, the gap net of covariates. The composition effect row reports $\hat{\delta} - \hat{\delta}_x$: the part of the gap a program could have predicted at the point of offer. Column (4) additionally absorbs the application set, which programs do not observe. Standard errors are clustered at the application set.

To estimate the moral hazard effect of unconditional offers, we use quasi-random variation in *exposure* to unconditional offers resulting from the idiosyncratic timing of unconditional offer adoption at the program level. Quasi-random variation in unconditional offer exposure allows us to separate the treatment effect of unconditional offers on test scores, which corresponds to moral hazard, from selection into take-up of unconditional offers, which corresponds to private information.

With this identifying variation, we can recover the causal effect of exposure on an outcome using a two-way fixed effects specification. Let $a(i)$ and $t(i)$ denote the application set and year of student i . Each student appears in a single year, so both are attributes of i , and we write a and t for them below. Define $Z_{at} \in \{0, 1\}$ as an indicator for whether any program in application set a gave more than 5% unconditional offers in year t , and let t_a^{first} be the first year in which set a is exposed. We then estimate

$$Y_i = \alpha_{a, P_{at}} + \gamma_1 Z_{at}^{\text{first}} + \gamma_2 Z_{at}^{\text{later}} + \lambda_t + X_i' \beta + \varepsilon_i, \quad (5)$$

where Y_i is the age-18 test score of student i , X_i a vector of academic and demographic controls (age-16 grades, teacher-predicted age-18 grades, socioeconomic quintile, and gender), and λ_t year fixed effects. The term $\alpha_{a, P_{at}}$ contains application set $\times$ post-first-exposure fixed effects, where $P_{at} = \mathbf{1}\{t > t_a^{\text{first}}\}$ indicates years after a set’s first exposure. The exposure indicator is split in two: $Z_{at}^{\text{first}} = \mathbf{1}\{t = t_a^{\text{first}}\}$, the first year an application set is exposed to unconditional offers, and

¹⁹See also Chiappori and Salanié (2000), which makes this point in the context of insurance markets.

$Z_{at}^{\text{later}} = Z_{at} \mathbf{1}\{t > t_a^{\text{first}}\}$, every subsequent year in which the set is exposed. Standard errors are clustered at the application set level.²⁰

Our primary coefficient of interest is γ_1 , the causal effect of first-year exposure to unconditional offers in one's application set. The coefficient γ_1 is causal under the assumption of strict exogeneity of treatment for first-year exposure. The core intuition for why this is likely to hold in this setting is that the first year of exposure comes as a "surprise" to students. Programs did not announce that they were using unconditional offers in advance, so students had little ability to predict which programs would introduce unconditional offers in that year. This assumption is consistent with the balance tests in Table A.3; in the year that unconditional offers began, application sets had no changes in observable characteristics in five of the six tests. In contrast, if students preferentially applied to programs that had given unconditional offers in the past, then γ_2 would not capture only the treatment effect of exposure, but also changes in applicant composition.

Figure A.1 shows the growth of both first-year and later exposure. Exposure spreads quickly once it begins: by 2018, 66.8% of students are exposed to unconditional offers through a program in their application set. From 2016 onward, most of those students are in sets exposed in an earlier year rather than newly exposed, and first-year exposure is the weaker treatment: 17.1% of first-year-exposed students receive an unconditional offer, against 31.0% of those exposed in later years.

Estimates from Equation (5) are reported in Table 5, and find strong evidence against the moral hazard hypothesis. Exposure to unconditional offers has a precisely estimated null effect on students' age-18 test scores. The precision of our estimates allows us to rule out declines of more than 0.010SD in final exam scores.

Table 5: Effect of exposure to unconditional offers

	(1) Receives an offer	(2) Uncond. take-up	(3) Achieved age-18 (z)
First exposure	0.166*** (0.004)	0.143*** (0.004)	0.005 (0.007)
Constant	0.014*** (0.002)	0.013*** (0.002)	0.259*** (0.002)
<i>N</i>	368,874	366,288	366,288

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate regression on an indicator for the first year in which any program in the student's application set makes unconditional offers. Column (1) is *receipt*: whether the student is made an unconditional offer at all. Column (2) is *take-up*: an unconditional offer ranked first or second, or a binding unconditional ranked first — a choice the student makes over the offers they hold. All columns include academic and demographic controls, year fixed effects, and application set $\times$ post-first-exposure fixed effects. The sample is years 2010–2019, main scheme applicants, programs averaging at least 20 offers per year over 2010–2012, with no program in the application set using unconditional offers in 2013 or earlier. Standard errors clustered at the application set, of which there are 97,730. Columns (2) and (3) are $\hat{\pi}$ and $\hat{\rho}$ in Table 6.

Estimates in Table 5 represent an intent-to-treat effect of exposure on test scores; however, as described in Equation (2), moral hazard is the average treatment effect on the treated. We can de-

²⁰We define application sets as unique combinations of {major $\times$ university} pairs applied to, aggregating across different programs within the same major at the same institution.

rive this quantity by interpreting [Table 5](#) as components of an instrumental variable specification, instrumenting take-up with exposure. Doing so requires the standard instrumental variable hypotheses. Relevance holds from column (2) of [Table 5](#), with $F = 1,555$. The exclusion restriction requires that exposure move achieved scores only through take-up. This is plausible: decliners would have no reason to change their behavior in response to having had such an offer available given that they declined. Independence follows from the strict exogeneity argument described above. As noted by [Bloom \(1984\)](#), given that unexposed students have close to zero take-up,²¹ the IV estimate corresponds to the average treatment effect on the treated — exactly the moral hazard term of interest. Under these assumptions, we find that

$$\text{Moral Hazard} = \mathbb{E}_Y[Y(1) - Y(0) \mid T = 1] = \frac{\rho}{\pi} = 0.033\text{SD}. \quad (6)$$

We conclude that take-up of unconditional offers does not cause students to achieve lower grades.

We now test for the presence of private information in our setting. Starting from [Equation \(4\)](#), express private information as the wedge

$$\text{Private Information} = \delta_X - \frac{\rho}{\pi}, \quad w \equiv \rho - \pi\delta_X = -\pi \text{Private Information}. \quad (7)$$

Testing for private information is then equivalent to testing $w > 0$. We report the results of this exercise in [Table 6](#). The wedge is positive and precisely estimated, $\hat{w} = 0.024$ with a bootstrapped standard error of 0.007, rejecting $w \leq 0$: the causal effect of take-up — moral hazard — cannot explain why takers underperform decliners.

From these results, we can bound the share of the conditional gap which can possibly be explained by moral hazard. Let $\theta = (\rho/\pi)/\delta_X$ be the share of the conditional taker–decliner achievement gap composed of moral hazard.²² We can bound θ by finding the smallest θ_0 at which $w(\theta_0) = \rho - \theta_0\pi\delta_X$ is significantly positive using a one-sided t -test. We find that $\theta \leq 0.346$: at most a third of the difference between takers and decliners is due to moral hazard, and the remainder is due to students’ private information.

These results are robust to alternate sample definitions, alternate thresholds for what counts as using unconditional offers, dropping the post-first-exposure block, and clustering at the least-selective university in the application set rather than at the application set, as reported in [Table A.4](#). As shown in [Table A.6](#), the decomposition is also robust to relaxing the assumption of no always-takers. Additionally, we estimate treatment effects of exposure on age-18 test scores using a quantile regression, and report results in [Table A.5](#); because results are insignificant across the ability distribution, we conclude that the null effect in [Table 5](#) is not driven by an increase in age-18 scores for high-achievers and a decrease for low-achievers, or vice versa.

²¹In [Table A.6](#), we relax this assumption as a robustness check.

²²The estimate of δ_X differs slightly between [Table 4](#) and [Table 6](#); the former is estimated on all unconditional offer recipients, while the latter only during the year of first exposure.

Table 6: Decomposing the taker–decliner achievement gap

	Estimate	SE	95% CI
<i>Panel A. Estimated inputs</i>			
$\hat{\pi}$, effect of exposure on take-up	0.143	(0.003)	[0.138, 0.151]
$\hat{\rho}$, effect of exposure on the achieved score (z)	0.005	(0.007)	[−0.010, 0.019]
$\hat{\delta}_X$, taker–decliner achievement gap net of covariates (z)	−0.134	(0.011)	[−0.156, −0.114]
<i>Panel B. Test for private information</i>			
$\hat{w} = \hat{\rho} - \hat{\pi}\hat{\delta}_X$	0.024	(0.007)	[0.009, 0.039]
<i>Panel C. Implied split of the gap</i>			
Moral hazard (ATT) = $\hat{\rho}/\hat{\pi}$	0.033	(0.048)	[−0.068, 0.128]
Private information = $\hat{\delta}_X - \hat{\rho}/\hat{\pi}$	−0.167	(0.049)	[−0.260, −0.064]
Moral-hazard share $\theta = (\hat{\rho}/\hat{\pi})/\hat{\delta}_X$	−0.244	(0.365)	≤ 0.346

Table reports decomposition of the taker–decliner achievement gap using the test described in [Section 3.1](#), yielding point estimates and bounds on θ , the share of the gap composed of moral hazard. The gap is estimated on 1,700,457 students, while exposure effects are estimated on 366,288 students in non-singleton application sets. Standard errors are from a cluster bootstrap over application sets with 200 replications. The last column reports two-sided 95% percentile intervals, except in the θ row, where it reports the one-sided 95% upper bound obtained by inverting $w(\theta_0)$: the largest share not rejected by the data. Robustness checks reported in [Table A.4](#).

3.2 Effects on matriculation and degrees

Unconditional offers do not impact students’ academic achievement. Do they affect which programs students attend? Using [Equation \(5\)](#), we estimate the causal effect of unconditional offer exposure on students’ trajectories through higher education.

We first examine results from university enrollment and degree attainment. Results are shown in panel (a) of [Figure 2](#) and column (1) of [Table 7](#). On net, unconditional offer exposure leads to an increase in university matriculation by 1.1pp, with the increase concentrated at middle-tercile universities. Recall that the effect of exposure on unconditional offer receipt is 16.6pp; again treating these results as an IV, we infer that students receiving an unconditional offer are 6.4pp more likely to matriculate at university.

Exposure also increases the quality of the program students attend. We define a program’s average achievement as the average age-18 test scores of students in that program during 2010–2013, before unconditional offers began. We then estimate [Equation \(5\)](#) with the dependent variable of program quality a student enrolls at, restricted to students that enroll in university. Among university enrollees, exposure to unconditional offers leads students to matriculate at programs whose average achievement is 0.008SD higher.

Finally, we turn to the long-term impacts of unconditional offer exposure on students’ educational trajectories. To do so, we estimate [Equation \(5\)](#) on years of university attended and degree completion by age 24, restricted to students for whom we observe data at age 24. Results are reported in panel (b) of [Figure 2](#) and columns (3) and (4) of [Table 7](#). Students exposed to unconditional offers spend an additional 0.052 years enrolled in university; however, this does not translate to a significant increase in university degree completion. Panel (b) of [Figure 2](#) shows that there is a significant decrease of degree completion at bottom-tercile universities; there is suggestive evi-

dence that this is offset by an increase in degree completion at middle- and top-tercile universities, though estimates are not significant at the 5% level.

Figure 2: Effect of unconditional offer exposure on matriculation and degree attainment, by university tercile. (a) Matriculation at ages 18–21, and ever; (b) holding an undergraduate degree by ages 21–24.

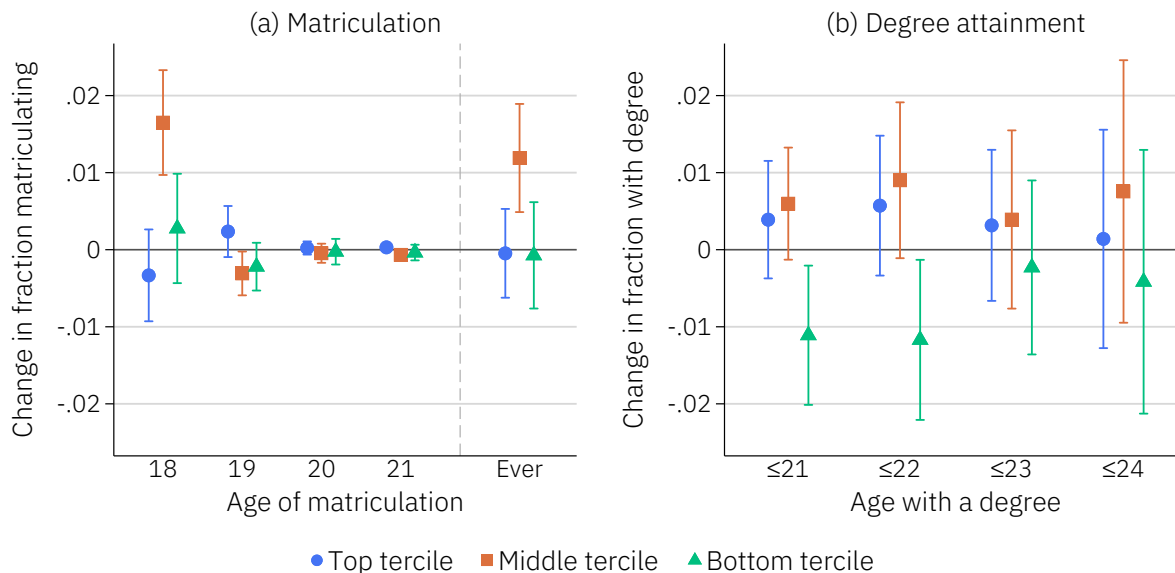


Table 7: Effect of unconditional offer exposure on matriculation, program quality, years enrolled, and degree attainment

	(1) Ever matriculates	(2) Program quality (z)	(3) Years enrolled by 24	(4) Degree by 24
First exposure	0.011*** (0.003)	0.008** (0.004)	0.052* (0.028)	0.005 (0.009)
Mean of dep. var.	0.926	0.301	3.044	0.798
<i>N</i>	368,870	296,990	126,810	126,810

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate estimation of Equation (5) on the outcome in its header, with the controls and fixed effects of Table 5, reporting the coefficient on first-year exposure; standard errors are clustered on the application set. Columns (1) and (4) are probabilities; column (2) is the average achieved z-score of the program a student matriculates at, as measured by average student achievement in 2010–2013, on a sample restricted to university matriculants; column (3) is years enrolled by age 24. *N* is rounded to the nearest 10.

4 A Structural Model of Application and Enrollment

The reduced-form results in Section 3 establish that unconditional offers had small positive effects on students' university matriculation with no impacts on their academic achievement. Policy recommendations depend on relative sizes of three additional factors: (1) the unconditional-offer premium, the value students place on an offer being unconditional, (2) the spillover effects of unconditional offers onto other students, and (3) how the match fares under alternative admissions systems.

We analyze these factors through a structural model of the UK university admissions market to estimate how students match to programs under counterfactual policies. The model is a four-stage game that mirrors the institutional timeline in Table 1; we estimate the preferences of students and programs for each other using techniques from Larroucau and Rios (2020), while students' private signals are modeled as in Kapor (2025).

Our reduced-form results allow us to design the structural model to identify the extent of private information. Because unconditional offers do not negatively impact students' grades, our model does not need to account for moral hazard. This allows direct identification of private information from the correlation of observed rank-order choices and realized grades. For example, if students who tend to rank more-selective programs also tend to outperform their expected age-18 grades, we infer that they possess private information about their academic ability.

4.1 Structural setup

The market proceeds in four stages. In Stage 1, each student applies to up to five programs, choosing a portfolio A_i . In Stage 2, programs can make both conditional and unconditional offers, resulting in each student holding an offer set $B_i \subseteq A_i$. In Stage 3, students receive a signal about their own ability and then rank up to two held offers $C_i \subseteq B_i \cup \{s_0\}$. In Stage 4, students take their age-18 examinations, and are admitted if they meet the relevant offer conditions. A student enrolls at her highest-ranked program that admits her, otherwise taking the outside option.

We describe each of these four stages in reverse chronological order. Let $\mathcal{I}$ index students and $\mathcal{S}$ index programs (university $\times$ major), with s_0 denoting the outside option of not enrolling, normalized to zero utility.

4.1.1 Stage 4: admission and enrollment

In the final stage, test scores are realized and students are admitted into programs.

The age-18 examinations reveal ability $r_i \sim N(0, \sigma_r^2)$. For a student i who ranked a conditional offer from program s , the offer converts to a place if and only if an augmented score clears the program's final bar,

$$z_i^{\text{offer}} \gamma_s^{\text{admit}} + \alpha_s r_i + \mu_i \geq \bar{\pi}_s^R, \quad (8)$$

where z_i^{offer} are pre-application characteristics observable to the program (age-16 grades, teacher-predicted age-18 grades, age-18 test composition, Low-SES, and indicators for missing data), γ_s^{admit} are final-stage weights on those characteristics, $\mu_i \sim N(0, 1)$ is an unobserved component of student quality that is common across programs, and $\bar{\pi}_s^R$ is the cutoff for acceptance. The common component μ_i captures application elements observed by programs but missing from the data, such as teacher references and a personal statement. It also enters the offer score (Equation (11)), reflecting that holistic quality informs both the offer and the final decision. An unconditional offer converts with certainty; a binding one converts only if ranked first. The student enrolls at her first-ranked program if admitted there, otherwise at her second-ranked program if admitted there, and otherwise takes the outside option s_0 .

4.1.2 Stage 3: ranking

In this stage, students receive a signal of their age-18 exam performance and then rank up to two programs.

Specifically, each student chooses an ordered portfolio (c_1, c_2) , with the programs chosen from their offer set B_i and their outside option s_0 . Unranked offers are forfeited.

Students' utility for an offer at Stage 3 is given by

$$U'_{is} = U_{is} + z'_i \lambda_i \cdot \mathbf{1}[s \text{ is an unconditional offer}] + \varepsilon_{is}, \quad (9)$$

where U_{is} is the stage-1 utility that student i has for program s , specified in [Section 4.1.4](#), $\varepsilon_{is} \sim N(0, \sigma_\varepsilon^2)$ is an i.i.d. idiosyncratic shock to utilities, and $z'_i \lambda_i$ is the unconditional-offer premium. We model the coefficient λ_i on the unconditional-offer premium with heterogeneity on the basis of gender and predicted age-18 grades. The shock ε_{is} is realized after the offer set is observed and represents what the student learns between applying and ranking, such as information from campus visits and updated assessments of fit.

Admission is uncertain because of students' uncertainty about r_i , their age-18 exam performance which has not yet been realized. However, students privately observe a signal of their test scores. In particular $r_i = \psi_i + e_i$, where the signal $\psi_i \sim N(0, \sigma_\psi^2)$ is known to the student but not to programs, and $e_i \sim N(0, \sigma_r^2 - \sigma_\psi^2)$ is independent of ψ_i . The share $\sigma_\psi^2 / \sigma_r^2$ governs how well a student can forecast her own age-18 test scores.

We assume that students have full knowledge of the offer process, including programs' parameters for admission and cutoffs, and so hold true beliefs about the likelihood of admittance²³ at each program. Students also learn about μ_i given their offer set B_i , and update their beliefs accordingly. Write $a_{is} \equiv \Pr(\text{admit } s \mid \psi_i, B_i)$ for the probability that [Equation \(8\)](#) holds for program s , given the student's signal and this posterior on μ_i , and $a_{i,c_2 \setminus c_1} \equiv \Pr(\text{admit } c_2, \text{ reject } c_1 \mid \psi_i, B_i)$.

The student chooses (c_1, c_2) to maximize the expected value of eventual enrollment,

$$\max_{c_1, c_2 \in B_i \cup \{s_0\}} V(c_1, c_2), \quad V(c_1, c_2) = a_{ic_1} U'_{ic_1} + a_{i,c_2 \setminus c_1} U'_{ic_2}. \quad (10)$$

4.1.3 Stage 2: offers

In this stage, programs decide to give conditional and unconditional offers to students.

Each program s decides which applicants receive offers, evaluating applicant i through a latent admissions score

$$\pi_{is} = z_i^{\text{offer}} \gamma_s + \mu_i, \quad (11)$$

where z_i^{offer} and μ_i are as in [Section 4.1.1](#) and γ_s are program-specific offer-stage weights, which may differ from the final-stage weights γ_s^{admit} because r_i is not yet available. Program s makes an offer if and only if $\pi_{is} \geq \bar{\pi}_s$, a program-specific threshold set to fill expected capacity. Equivalently, an offer from s is the event $\mu_i \geq L_{is} \equiv \bar{\pi}_s - z_i^{\text{offer}} \gamma_s$.

Through μ_i , a student's offers are positively correlated across programs. A student who receives

²³As discussed in [Agarwal and Somaini \(2018\)](#), utility and beliefs about admission probability cannot be jointly estimated, so we pin the latter at its true value.

an offer from a program that is highly selective on μ_i will also receive offers from all programs requiring a lower value of μ_i .

We also allow programs to set a cutoff $\bar{\pi}_s^U$ for unconditional offers, so that a student receives an unconditional offer if and only if $\pi_{is} \geq \bar{\pi}_s^U$. We assume $\bar{\pi}_s^U > \bar{\pi}_s$, so that the highest-scoring subset of students with offers receive an unconditional offer.

4.1.4 Stage 1: initial applications, portfolio choice

In this stage, students form their application portfolio and apply to up to five programs.

Student i 's indirect utility from enrolling at program s is

$$U_{is} = \delta_s - d_{is} + \beta_i^c \text{close}_{is} + \beta^m m_{is} + v_{is}, \quad v_{is} \sim N(0, \sigma_v^2), \quad (12)$$

where the idiosyncratic shock v_{is} is drawn independently across student–program pairs and $U_{i0} = 0$. Preferences rely on program fixed-effects δ_s , motivated by the fact that our core question is not which program characteristics students value, but how much they value them relative to the unconditional-offer premium.

Three terms are functions of student locations: d_{is} , the distance in kilometers between a student's postcode and the institution; $\text{close}_{is} = \mathbb{1}[d_{is} < 40]$, for being within commuting range; and m_{is} , for student and program sharing a UK region.²⁴ The coefficient on distance is normalized to -0.01 , so that a unit of utility is the value of 100km of avoided travel at the margin.

We allow β_i^c to vary across students, capturing heterogeneity in the value of attending university near to one's home. We model this heterogeneity as a function of students' socioeconomic status:

$$\beta_i^c = \bar{\beta}^c + \Gamma^c \text{Low-SES}_i + \eta_i^c, \quad \eta_i^c \sim N(0, \sigma_\eta^2), \quad (13)$$

where Low-SES_i marks students from the two most deprived socioeconomic quintiles, who are likely to be more sensitive to being within commuting distance.

When applying, students know z_i^{offer} but not μ_i , so they face marginal offer probabilities of

$$p_{is} = \Pr[\text{offer from } s \mid z_i^{\text{offer}}] = \Phi(z_i^{\text{offer}} \gamma_s - \bar{\pi}_s). \quad (14)$$

Student i chooses a portfolio of at most five programs to which to apply, $A_i \subseteq \mathcal{S}$. Let B_i be her offer set and C_i her ranked programs. Then, she chooses A_i such that

$$A_i = \arg \max_{A \subseteq \mathcal{S}, |A| \leq 5} V(A), \quad V(A) = \mathbb{E}_B[V_i(C_i(B))], \quad (15)$$

where $C_i(B)$ is the optimal ranking choice for i given offer set B ; this is the ranking problem of [Section 4.1.2](#). The key detail is that students are uncertain about their offers and eventual admission when they apply.

²⁴The 12 International Territorial Level 1 regions, as defined by the Office for National Statistics.

4.2 Model assumptions

The model as stated involves a high-dimensional optimization at the application stage and a rich correlation structure across programs. We impose assumptions that discipline student behavior and beliefs, making our model tractable from the observed data. We then draw out the implications of our assumptions, used by our estimator.

Assumption 1 (Portfolio evaluation). When choosing a portfolio, students value it as though all held offers can be ranked. Further, they do not anticipate the arrival of their private signal ψ_i nor the Stage-3 idiosyncratic shock.

This makes the application-stage problem tractable without collapsing conversion risk by reducing Equation (15) from a dynamic problem to a single optimization. A fully rational model would involve students reasoning through all possible scenarios of offer sets, grade outcomes, and preference shocks.

Assumption 2 (Correlated offers). A student's offer events are comonotone across programs, generated by the common holistic component μ_i in Equation (11) rather than being independent across programs.

This is natural in the UK context, where admission depends heavily on academic grades; in the data, 92.5% of students who receive an offer from a given program receive offers from all less-selective programs they apply to.

These two assumptions imply that portfolio utility can be expressed as a linear combination of admission probabilities and utilities, granting us traction on students' stage-1 portfolio problem.

Implication 1 (Portfolio valuation). Under Assumption 1 and Assumption 2, index the $K = |A_i| \leq 5$ programs in A_i so that $q_{a_1}^i \leq q_{a_2}^i \leq \dots \leq q_{a_K}^i$, where q_{is} is the probability that s makes student i an offer and admits her, and set $q_{a_0}^i \equiv 0$. Then the expected utility of A_i is

$$V(A_i) = \sum_{B \subseteq A_i} P_i(B) \max_{b \in B} U_{ib} = \sum_{k=1}^K (q_{a_k}^i - q_{a_{k-1}}^i) \max_{j \geq k} U_i(a_j),$$

where $P_i(B)$ is the probability that the programs admitting student i are exactly those in B .

The first equality follows from Assumption 1. The second follows from Assumption 2: a student admitted to a_k is also admitted to every a_j with $j > k$, so with probability $q_{a_k}^i - q_{a_{k-1}}^i$ her admitted set is $\{a_k, \dots, a_K\}$. Together they make $V(A_i)$ linear in admission probabilities, and under Implication 2 below the inner maximum is $U_i(a_k)$.

The relevant probability is the *total* chance of admission q_{is} , not the offer probability alone: a program with a low offer threshold but a high admission threshold may not be a safe choice. Because μ_i enters both Equation (11) and Equation (8), receiving an offer and meeting its condition are positively dependent, so students compute q_{is} as a joint probability rather than the product of the two marginals:

$$q_{is} = p_{is} [g_{is} + (1 - g_{is}) \Pr(\text{admit } s \mid \text{offer from } s)],$$

where g_{is} is the probability that an offer from s is unconditional, and the conditional admission probability integrates over μ_i truncated at the offer threshold and over r_i given z_i^{offer} .

Implication 2 (No dominated programs). For a chosen portfolio A_i , where $q_{a_1}^i \leq q_{a_2}^i \leq \dots \leq q_{a_5}^i$ are the ordered admission probabilities of programs in A_i , we have that

$$U_i(a_1) \geq U_i(a_2) \geq \dots \geq U_i(a_5).$$

This is equivalently stated as a “no dominance” condition: if, for some programs s and s' , we have that $q_{is} < q_{is'}$ and $U_{is} < U_{is'}$, then student i will never apply to s .

This implication imposes a utility ordering on the programs a student applied to even though no ranking among them is observed. It follows from [Implication 1](#) and portfolio optimality.

Assumption 3 (No moral hazard). Realized performance r_i is independent of B_i , conditional on A_i and z_i^{offer} .

In line with our results from [Section 3](#), we assume that students do not change their effort in response to the offers they hold.²⁵

4.3 Estimation procedure

We estimate the model in three blocks. We recover odds of offers and admissions through the supply side, estimate student utilities using initial applications, then use Stage 3 to recover the coefficient λ_i on the unconditional-offer premium and the signal precision σ_ψ^2 . The three blocks are estimated sequentially rather than jointly; [Appendix F.6](#) discusses the validity of this sequential estimator, together with convergence diagnostics. Additional details are available in [Appendix F](#).

4.3.1 Supply side

We estimate the likelihood of receiving an offer and the likelihood of converting that offer to a place separately. Note the former is observed for all applying students, while the latter is observed only for students who rank the program and are “at-risk”. We use Firth-penalized probit to estimate the offer and conversion probabilities, which handles the perfect separation arising at the most and least selective programs. While μ results in a common shock, this model is still correctly specified because μ is integrated out of all marginal probabilities. Let these probabilities be $\hat{p}_{is}$ and $\hat{a}_{is}$, respectively:

$$\hat{a}_{is} = \Phi(z_i^{\text{offer}'} \hat{\gamma}_s^{\text{admit}} + \hat{\alpha}_s \hat{r}_i + \hat{c}_s^R), \quad \hat{p}_{is} = \Phi(z_i^{\text{offer}'} \hat{\gamma}_s + \hat{c}_s),$$

where the fitted intercepts $\hat{c}_s = -\hat{\pi}_s$ and $\hat{c}_s^R = -\hat{\pi}_s^R$ recover the offer and admission cutoffs of [Equation \(14\)](#) and [Equation \(8\)](#).

Students’ Stage-1 admission probability combines the two as in [Implication 1](#):

$$\hat{q}_{is} = \hat{p}_{is} [\hat{g}_{is} + (1 - \hat{g}_{is}) \hat{a}_{is}^o],$$

²⁵Moral hazard, if present, would make a stronger case for post-qualification admissions in [Section 4.6](#).

where $\hat{g}_{is}$ is the probability that an offer from s is unconditional, implied by s ’s unconditional-offer cutoff, and $\hat{a}_{is}^o$ is the conversion probability conditional on holding the offer. Because the offer and conversion indices share μ_i , we compute $\hat{a}_{is}^o$ from their bivariate normal distribution, with correlation $1/\sqrt{1 + \hat{\alpha}_s^2 \sigma_r^2}$, rather than setting it to $\hat{a}_{is}$.

4.3.2 Demand side, Stage 1

Following [Larroucau and Rios \(2020\)](#), optimal portfolios must satisfy a set of “one-shot swap” (OSS) inequalities:

$$V(A_i; q_i, U_i) \geq V((A_i \setminus \{s\}) \cup \{\ell\}; q_i, U_i) \quad \text{for all } s \in A_i, \ell \notin A_i, \quad (16)$$

where the swap removes s from the portfolio and adds ℓ . These inequalities exactly define the optimal portfolio when admissions are independent; effectively, they require that switching one chosen program for an unchosen program leads to lower utility.²⁶ However, under correlated admissions, these inequalities are necessary, but not sufficient, for optimality. The exact optimality conditions are the dynamic-programming inequalities of [Ali and Shorrer \(2025\)](#), which take into consideration the “continuation value” of programs — rejection from one program is informative about acceptance at another. However, the interconnected nature of these inequalities makes them computationally intractable for our application; the OSS inequalities act as a tractable approximation of optimality. In [Appendix E](#), we characterize the gap between the true optimality conditions and those in [Equation \(16\)](#); in our setting, given that the residual correlation in admissions likelihood after conditioning on observables is 0.06, we infer that the OSS inequalities are a good approximation to the full dynamic-programming inequalities.²⁷

The observed portfolio admits no closed-form choice probability; as such, we use Markov chain Monte Carlo algorithms to estimate Stage-1 utilities for every student-program combination. Student utilities must satisfy inequalities $\mathcal{L}_i > 0$, where $\mathcal{L}_i$ consists of three blocks of inequalities: a non-negativity constraint, $U_{is} > 0$ for all $s \in A_i$; the ranking constraint implied by [Implication 2](#); and the one-shot swap inequalities described in [Equation \(16\)](#). We draw from the joint posterior of the Stage-1 parameters and the utilities $\{U_i\}$ by Gibbs sampling; [Appendix F.2](#) gives the blocking, and [Appendix F.7](#) gives the priors and run lengths.

We treat q_{is} as latent and resample it with the utilities, proposing the offer-probit coefficient vector from its asymptotic distribution and accepting if the observed portfolio remains optimal. The conversion-probit component of q_{is} is held fixed at its point estimate. To account for possible mis-specification of believed admissions odds, we only use the OSS constraints and non-negativity constraints to define $\mathcal{L}_i$ when redrawing q .²⁸

²⁶[Larroucau and Rios \(2020\)](#) show that the outcome of the [Chade and Smith \(2006\)](#) “simultaneous search” algorithm is the unique portfolio identified by this set of inequalities.

²⁷A separate interpretation of our methodology is that by using the OSS inequalities, rather than the full dynamic-programming inequalities, our parameters are set-identified, rather than point-identified. In practice, given the standard errors of our estimates, the identified set is a good approximation for point-identification.

²⁸If we also required the ranking constraints, then all students hold true beliefs about the relative selectivity of programs. By [Implication 2](#), all students thus hold the same utility ordering across all programs. Thus, we relax the ranking constraint in this step to allow for a more flexible utility specification.

4.3.3 Demand side, Stage 3

The ranking stage takes the Stage-1 utilities estimated in [Section 4.3.2](#) as given. We estimate the Stage-3 parameters $(\sigma_\varepsilon^2, \lambda_i, \sigma_\psi^2)$ again using a Gibbs sampler subject to a system of inequalities. That student i ranked (c_1, c_2) rather than any alternative pair in B_i implies

$$a_{ic_1} U'_{ic_1} + a_{i,c_2 \setminus c_1} U'_{ic_2} \geq a_{ic'_1} U'_{ic'_1} + a_{i,c'_2 \setminus c'_1} U'_{ic'_2}, \quad (17)$$

with U'_{is} defined as in [Equation \(9\)](#). There are at most 26 such inequalities. As in Stage 1, we can redraw U'_{is} for every student-program combination subject to these constraints.

The signal ψ_i is latent and augmented alongside the ranking shocks: each sweep draws ψ_i given the observed first and second rankings and the take-up decision, then draws σ_ψ^2 given $\{\psi_i\}$ and realized $\{r_i\}$, with conversion probabilities recomputed under the student's posterior wherever they enter [Equation \(17\)](#). The ψ_i draw is a Metropolis step within the block; every other draw is conjugate.

4.3.4 Sample

We utilize the same sample restrictions as in [Section 3](#), with the additional restriction that students must have applied only to mathematics programs. The supply-side probits are fit on the full applicant population, for additional precision. The Stage-1 sampler is run for 50,000 iterations (5,000 discarded as burn-in) and the Stage-3 sampler for 250,000 iterations (25,000 discarded); [Appendix F.7](#) gives priors and further implementation detail.

4.4 Identification

Supply-side identification is standard: each per-program probit is identified by the excluded instrument of distance, following [Kapor \(2025\)](#). We focus on the demand side, where the argument follows [Larroucau and Rios \(2020\)](#), [Agarwal and Somaini \(2020\)](#), and [Kapor \(2025\)](#).

We pin the scale of students' preferences by fixing the distance coefficient to -0.01 , so that every parameter is measured relative to 100km of avoided travel. Utility's location is pinned by the outside option, normalized to zero.

Student preferences are identified by two objects. As distance is excluded from programs' utility over students, distance acts as a special regressor ([Agarwal and Somaini, 2020](#)), anchoring its units free of supply-side contamination. Identification of the remaining utility parameters is derived from the OSS inequalities ([Larroucau and Rios, 2020](#)); here, identification comes from year-to-year variation in cutoffs and grade requirements, which "trace out" regions of utility.

Private information is identified from differences in Stage-3 behavior among students holding the same menu of offers with the same observable characteristics. The data provides three margins of variation that identify σ_ψ^2 : the first is the take-up rate of unconditional offers, the second is the achievement gap between takers and decliners, and the third is the selectivity gap between students' first and second choice.

The unconditional-offer premium is identified by variation in the availability of unconditional offers that is unrelated to student preferences. The argument here follows the same argument as in [Section 3](#): the supply-side rollout of unconditional offers is plausibly exogenous to student

characteristics, and so comparing exposed cohorts to unexposed cohorts identifies the coefficient λ_i .

4.5 Results

Table 8: Structural parameter estimates

<i>Preferences over programs</i>			<i>Unconditional-offer premium coefficients λ_i</i>		
	Estimate	SD		Estimate	SD
Distance (per km)	−0.01	(anchor)	At mean predicted ability	0.119	0.204
Can live at home	−0.409	0.030	Gradient, per SD of score	−1.181	0.180
× Low-SES Γ^c	1.732	0.054	Female	1.062	0.309
SD across students σ_η	2.591	0.023	Ranking shock σ_ε	4.820	0.041
Same region	0.712	0.019			
Constant	−1.303	0.022			
Idiosyncratic shock σ_v	2.917	0.011	<i>Information</i>		
			Private-signal variance σ_ψ^2	0.221	0.006
			Variance of age-18 scores σ_r^2	0.331	—
			Share privately observed	0.67	—

Posterior means and posterior standard deviations from the Stage-1 and Stage-3 samplers of [Section 4.3.2](#) and [Section 4.3.3](#), for students applying only to mathematics programs. Distance is normalized to −0.01, so utility is in hundreds of kilometers. Only live-at-home carries a random coefficient and a Low-SES interaction. σ_η , σ_ν , σ_ε are square roots of sampled variances, with delta-method SDs; the posterior SD is recovered as $\text{MCSE} \times \sqrt{\text{ESS}}$. Rows without an SD are fixed or derived quantities for which the sampler reports no posterior standard deviation. Convergence statistics are reported in [Table A.7](#); notably, the convergence diagnostics for σ_ψ^2 indicate that its SD is likely understated.

[Table 8](#) reports the Stage-1 parameters. Because the distance coefficient is normalized to −0.01, every parameter is a willingness to travel, and we report all of them in kilometers.

High-SES students prefer to live away from home: the mean home coefficient is −41km, precisely estimated. However, this figure masks heterogeneity, both idiosyncratic and related to SES. The SD of η_i^c is 259km, and Low-SES students value living at home at an average of 132km of distance. Students additionally prefer attending a university in the same region of the UK as their home, valuing region match at 71km.

The unconditional-offer premium is steeply decreasing in ability and correlates with gender. At mean predicted ability a male student values one at just 12km (SD 20.4km, not distinguishable from zero), while female students value them at 118km. With a gradient of −118km per standard deviation of predicted ability, unconditional offers are not valued by the median male student in mathematics, and not valued by higher-achieving women.

While the median student does not value the certainty provided by unconditional offers, they still can benefit through the channel of private information, as students with negative signals will preferentially take unconditional offers. We estimate that age-18 achievement has a variance of $\sigma_r^2 = 0.331$, and that students privately observe two-thirds of it ($\sigma_\psi^2/\sigma_r^2 = 0.67$). This result aligns with the results in [Section 3](#): as students hold much private information about their own ability, they can advantageously self-select into taking unconditional offers on the basis of private signals.

4.6 Counterfactuals

The estimated parameters let us re-solve the market under counterfactual designs and compare the resulting matches and welfare. We consider three application system designs: a full ban on unconditional offers, analogous to the system in 2010–2013; unconditional offer shares held at each program’s 2018 level, the year with the most unconditional offers in this sample; and post-qualification admissions, in which applications are made after age-18 results are released and r_i is common knowledge. Post-qualification admissions are a particular counterfactual of interest; it is the only counterfactual in which students’ private signal plays no role in their admissions choices, as r_i has been revealed publicly. Under these counterfactuals, students gain the unconditional-offer premium only under the “2018 level” counterfactual.

Details on each design are given in [Appendix F.8](#). We report welfare relative to the ban — the design in which no student receives an unconditional offer — in kilometer-equivalents, and we include unplaced students, valued at the outside option (normalized to zero), so that designs which place different numbers of students remain comparable.

We report results from our counterfactual simulations in [Table 9](#) and [Table 10](#). The latter contains estimates of enrollment and average student ability at the program level, split by the quartiles of programs’ average achievement in 2010–2013.

Table 9: Welfare under counterfactual application system designs

	Uncond. offer in 2018		By achieved attainment				
Design	Yes	No	Q1	Q2	Q3	Q4	All students
<i>Panel A. Total average student utility</i>							
Unconditional shares of 2018	+0.345	−0.217	+0.086	−0.220	−0.190	−0.047	−0.093
Post-qualification admissions	+1.791	+1.629	−0.050	+1.817	+2.359	+2.535	+1.665
<i>Panel B. Utility from the unconditional-offer premium</i>							
Unconditional shares of 2018	+0.277	—	+0.235	+0.028	−0.003	−0.014	+0.061
<i>Panel C. Students placed</i>							
Unconditional shares of 2018	+278	−388	+39	−66	−63	−20	−110
Post-qualification admissions	+113	+401	−197	+357	+292	+63	+514
Students	6,926	24,480	7,852	7,851	7,851	7,852	31,406

Each cell is the mean difference from the ban on unconditional offers, per simulated market: in student utility, measured in hundreds of kilometers (Panels A and B), or in the number of students placed (Panel C). Unplaced students have utility 0. Panel A includes the utility from the match and the unconditional-offer premium $z'_i \lambda_i$ for students placed on an unconditional offer; Panel B reports the premium alone. Q1–Q4 are quartiles of achieved age-18 attainment. The first two columns split students by whether they receive an unconditional offer under 2018 shares; each student keeps that group in every design. Under the ban, mean utility is 13.34 and 25,202 students are placed. Differences are averaged over 30 simulations, paired within simulation. Details on counterfactuals are discussed in [Appendix F.8](#).

[Table 9](#) shows each design’s welfare change split by whether students received an unconditional offer in the 2018 counterfactual, split by academic quartile, and in aggregate. Each cell is a change relative to the ban. Panel A reports mean student utility in hundreds of kilometers, Panel B the part of that utility that comes from the unconditional-offer premium, and Panel C the change in the number of students placed.

Unconditional offers benefit recipients, but redistribute utility and placements from high-ability

students to low-ability students. On net, students lose utility equivalent to 9.3km (−0.7%); recipients of unconditional offers gain 34.5km in average utility, but non-recipients are hurt by 21.7km. The academic quartile split shows a similar pattern: the lowest quartile benefits the most from unconditional offers, hurting higher-achieving students, particularly the second-lowest quartile. The losses for higher-achieving students come from misallocation (Panel A) and reduced placement rates (Panel C), while low-achievers primarily benefit from the unconditional-offer premium (Panel B). The intuition behind this result is that low-achieving students, knowing that they are unlikely to do well, are the ones who choose to take an unconditional offer, while high-achieving students are more likely to forgo it in favor of pursuing admission based on their test scores. After the exams, low-achieving students had already secured a place of admission, and so higher-achieving students have fewer spots to fill. One interpretation of this result is that unconditional offers remove slack from the market, which reduces market efficiency.

In contrast, post-qualification admissions benefit high-achieving students, leading to overall welfare gains. In aggregate, students gain 166.5km in utility (12.5%) under post-qualification admissions. Benefits under post-qualification admissions primarily occur through improved allocation. Students in the top quartile of achievement have little change in their placement rate, but gain because they can attend more-preferred programs.²⁹ Students in the second-lowest quartile of achievement also benefit: their university matriculation rate increases by 4.5pp. One way this can occur is if these students are indistinguishable from the lowest-quartile, and both do not receive offers; if admission runs after the release of age-18 exam scores, their performance helps them earn a university seat.

Table 10: Where students enroll, by program

	Uncond. offers in 2018		By program selectivity				
Design	Adopters	Non-adopters	Q1	Q2	Q3	Q4	All programs
<i>Panel A. Students enrolled</i>							
Unconditional shares of 2018	+145	−255	−11	−17	−8	−74	−110
Post-qualification admissions	+74	+440	+122	+268	+157	−33	+514
<i>Panel B. Mean age-18 scores of incoming students (SD)</i>							
Unconditional shares of 2018	−0.068	+0.010	+0.002	−0.028	−0.013	+0.008	−0.004
Post-qualification admissions	+0.084	+0.002	+0.108	+0.196	+0.060	−0.073	+0.012
Programs	16	44	15	15	15	15	60
Mean enrolled under the ban	3,127	22,075	2,812	3,387	7,099	11,904	25,202
Mean attainment under the ban (SD)	0.577	1.005	−0.068	0.393	0.984	1.332	0.952

Each cell is the mean difference from the ban on unconditional offers, per simulated market, in the number of students a program enrolls (Panel A) and in the average achieved age-18 scores of its incoming class (Panel B). Adopters are programs with a positive unconditional-offer share in 2018. Q1–Q4 are quartiles of programs' average incoming scores in 2010–2013 (Q4 is the most selective). Differences are averaged over 30 simulations, paired within simulation. Details on counterfactuals are discussed in [Appendix F.8](#).

Even though post-qualification admissions increase allocative efficiency for students, such a policy cannot occur unless universities are willing to adopt it. In [Table 10](#), we show how coun-

²⁹Students may not have even applied to these programs at baseline, as they have not received their Stage-3 idiosyncratic shock. Under post-qualification admissions, students' behavior can reflect these preferences.

terfactual designs impact program enrollment and average incoming student ability. We report these changes split by unconditional offer adoption status in 2018, quartiles of average achievement between 2010 and 2013, and in aggregate. We first note that programs across the selectivity distribution experience a decline in enrollment when unconditional offers are introduced, with minor changes in achievement as well. This evidence is suggestive of why universities may have collectively banned unconditional offers in 2021: they decreased enrollment and lower-achieving students were more likely to attend university.

In contrast, post-qualification admissions benefit most programs, but not the top quartile. The intuition is that some high-achieving students, after receiving shocks to their per-program utilities, can now re-optimize and attend their new most-preferred program, which does not necessarily belong to the top quartile of programs. Their enrollment losses are small — only 0.3% — but they do experience a net decrease in incoming student achievement of 0.073SD. Less-selective programs are the ones that benefit from increased enrollment and increased student quality, particularly the second-lowest quartile of programs. Post-qualification admissions, while largely beneficial for both students and programs, have some detrimental effects for the most-selective programs.

5 Conclusion

The design of admissions systems matters. In the context of UK university admissions, unconditional offers increase students' matriculation rates with no negative impact on their achievement, a finding that goes against the prevailing narratives. Using this finding, we develop a model to evaluate the total welfare impacts of such offers, accounting for student heterogeneity, private information, and a preference for certainty in admissions. In aggregate, unconditional offers lead to a net loss in student welfare and transfer utility from high-achieving students to low-achieving students. We then show that post-qualification admissions can improve student welfare by 12.5%, with benefits accruing to higher-achieving students.

The UK Department for Education considered moving to post-qualification admissions in 2022, but ultimately decided against it over many concerns, particularly the compressed timeline that universities would need to admit students. Our findings suggest that this debate should be re-opened given the large benefits to students; other recent media has also suggested this given the rise of artificial intelligence ([Harvey-Brewin, 2026](#)). However, our model suggests that overall welfare could be improved by helping students make informed choices at the application stage. This avenue is far less disruptive to students and could potentially achieve similar welfare gains without the logistical challenges of post-qualification admissions.

Our results suggest several avenues for future research. How much do our results extrapolate to other non-education settings, such as labor market unraveling?³⁰ How strong is private information in these settings? Does private information act as a force against the unraveling of these markets? Development and estimation of dynamic best-response models incorporating private information could shed light on these issues.

³⁰Empirical work has documented unraveling in several different markets: [Niederle and Roth \(2003\)](#) and [Herrmann et al. \(2022\)](#) in medical fellowship hiring, [Fréchette, Roth and Ünver \(2007\)](#) in college football scheduling, and [Avery et al. \(2007\)](#) in law clerk hiring.

On a separate note, private information about ability is necessary for the strong selection into take-up that is present in our setting. A recent literature suggests that parents often have inaccurate beliefs about children's ability ([Dizon-Ross, 2019](#); [Wang, Cheng and Smyth, 2022](#)) — so where does students' private information come from? Perhaps there is a wedge between parent and child perceptions of ability, or perhaps the structure of secondary education in the UK has allowed students to gain stronger signals of their own performance. Identifying reasons behind and ways to close this gap could help improve the efficiency of familial education investments.

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A Supplementary Information

Table A.1: Characteristics of programs that adopt unconditional offers

Panel A. Program types

	(1) Top tercile	(2) Middle tercile	(3) Bottom tercile	(4) STEM	(5) Arts	(6) Enrollment
Uses unconditional offers	−0.389*** (0.012)	0.082*** (0.013)	0.307*** (0.013)	−0.112*** (0.013)	0.035*** (0.011)	−1.927** (0.975)
Constant	0.477*** (0.010)	0.271*** (0.009)	0.252*** (0.009)	0.398*** (0.010)	0.174*** (0.008)	32.19*** (0.745)

Panel B. Baseline program selectivity, enrolling class characteristics

	(7) Offer rate	(8) Rate ranked first	(9) Age-16 English (z)	(10) Age-16 math (z)	(11) Teacher- predicted age-18 (z)	(12) Achieved age-18 (z)
Uses unconditional offers	0.092*** (0.006)	−0.065*** (0.004)	−0.355*** (0.013)	−0.457*** (0.015)	−0.538*** (0.016)	−0.530*** (0.016)
Constant	0.758*** (0.005)	0.297*** (0.003)	0.298*** (0.011)	0.413*** (0.012)	0.038*** (0.014)	0.173*** (0.013)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate regression of the named program characteristic, measured in the baseline year, on an indicator for whether more than 5% of a program's offers are unconditional. Observations are programs; the sample is programs averaging at least 20 offers per year over 2010–2012 that did not use unconditional offers in 2013 or earlier. $N = 4,990$, except in columns (9)–(12), where it is 4,976. Standard errors in parentheses.

Table A.2: Characteristics of unconditional offer recipients and takers, by timing of exposure

	Receiving an unconditional offer			Take-up, among offered		
	(1) No offer	(2) Received offer	(3) Difference	(4) Decliner	(5) Taker	(6) Difference
<i>Panel A. First-year exposure</i>						
Age-16 English (z)	−0.007 (0.003)	0.086 (0.006)	0.093*** (0.007)	0.129 (0.013)	0.076 (0.006)	−0.053*** (0.015)
Age-16 math (z)	0.022 (0.003)	0.062 (0.007)	0.040*** (0.008)	0.114 (0.015)	0.050 (0.007)	−0.064*** (0.017)
Teacher-predicted age-18 (z)	−0.299 (0.003)	0.101 (0.006)	0.400*** (0.007)	0.251 (0.014)	0.068 (0.007)	−0.183*** (0.015)
Achieved age-18 (z)	−0.282 (0.003)	−0.037 (0.007)	0.245*** (0.008)	0.191 (0.014)	−0.089 (0.007)	−0.280*** (0.016)
<i>Panel B. Later-year exposure</i>						
Age-16 English (z)	−0.083 (0.002)	−0.113 (0.003)	−0.030*** (0.004)	−0.091 (0.007)	−0.119 (0.004)	−0.028*** (0.006)
Age-16 math (z)	−0.106 (0.003)	−0.206 (0.003)	−0.100*** (0.005)	−0.175 (0.008)	−0.213 (0.004)	−0.039*** (0.007)
Teacher-predicted age-18 (z)	−0.442 (0.003)	−0.227 (0.003)	0.215*** (0.005)	−0.151 (0.008)	−0.243 (0.004)	−0.092*** (0.007)
Achieved age-18 (z)	−0.395 (0.003)	−0.331 (0.003)	0.064*** (0.005)	−0.154 (0.008)	−0.368 (0.005)	−0.214*** (0.007)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Columns are defined as in Table 2. Panel A repeats the first-exposure figures; Panel B reports the same quantities for students in a year after the first in which any program in their application set makes unconditional offers.

Table A.3: Balance of student characteristics at first exposure to unconditional offers

	(1) Age-16 English (z)	(2) Age-16 math (z)	(3) Independent school	(4) Female	(5) Low-SES	(6) Predicted age-18 score (z)
First year exposed	0.001 (0.007)	0.007 (0.007)	0.041 (0.026)	−0.001 (0.005)	−0.002 (0.004)	0.022*** (0.008)
Later years exposed	−0.007 (0.014)	0.014 (0.016)	−0.044 (0.055)	−0.004 (0.008)	0.011 (0.010)	0.017 (0.017)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column re-estimates Equation (5) with a student characteristic measured before application as the outcome, and without the academic and demographic controls. Indicators for missing age-16 and teacher-predicted scores are estimated alongside and not reported. Observations are students; the sample is years 2010–2019, main-scheme applicants whose application sets contain no program that used unconditional offers in 2013 or earlier and whose programs each average at least 20 offers per year over 2010–2012. $N = 370,973$, except in column (5), where it is 368,907. Standard errors are clustered at the application set level.

Figure A.1: Time series of exposure to unconditional offers. (a) Share of students in each exposure group over time; (b) share of students with an unconditional offer, split by exposure group.

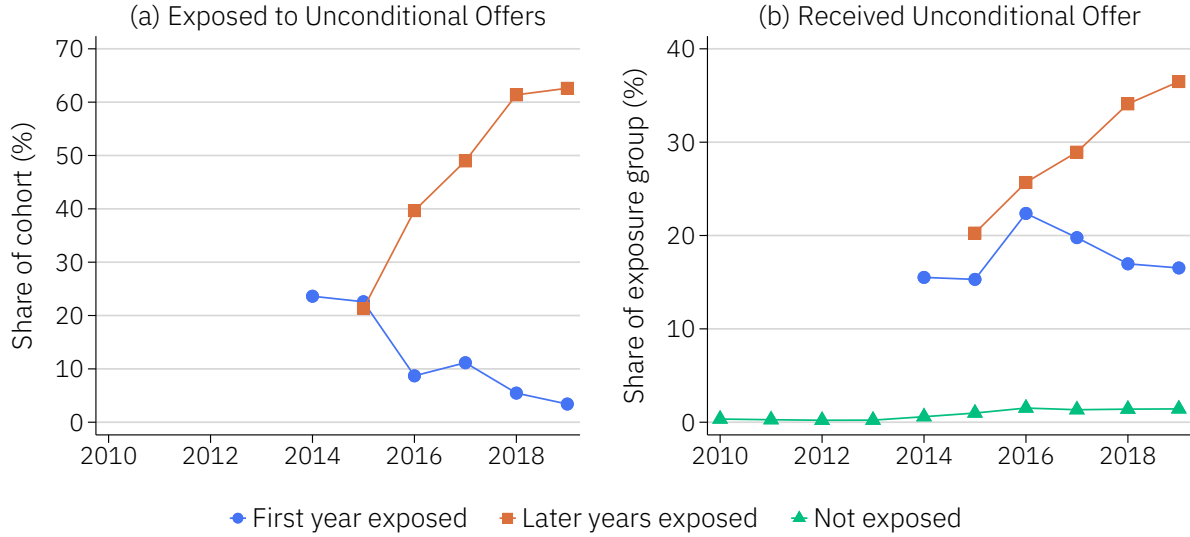


Table A.4: Robustness of the decomposition

	Main	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\pi}$	0.143	0.143	0.177	0.141	0.146	0.144	0.144	0.149	0.145
$\hat{\rho}$	0.005	0.005	0.001	0.004	0.005	0.004	0.003	0.002	0.004
$\hat{\delta}_x$	-0.134	-0.134	-0.122	-0.128	-0.133	-0.138	-0.139	-0.134	-0.136
$\hat{w}$	0.024 (0.007)	0.024 (0.007)	0.022 (0.008)	0.022 (0.009)	0.025 (0.007)	0.024 (0.007)	0.023 (0.007)	0.022 (0.008)	0.024 (0.007)
t	3.43	3.50	2.83	2.29	3.70	3.52	3.27	2.70	3.43
ATT	0.033	0.033	0.003	0.026	0.036	0.026	0.019	0.012	0.026
PI	-0.167	-0.167	-0.125	-0.154	-0.169	-0.165	-0.159	-0.146	-0.162
$\hat{\theta}$	-0.244	-0.244	-0.023	-0.205	-0.271	-0.191	-0.140	-0.088	-0.194
θ cap	0.346	0.327	0.561	0.650	0.281	0.352	0.420	0.563	0.365
Clusters	97,730	97,730	97,004	59,709	103,379	100,998	93,898	5,007	81,892

$\hat{\pi}$ is the first stage, $\hat{\rho}$ the reduced form and $\hat{\delta}_x$ the taker–decliner achievement gap; $\hat{w} = \hat{\rho} - \hat{\pi}\hat{\delta}_x$, with its standard error in parentheses and t its t -statistic; $ATT = \hat{\rho}/\hat{\pi}$ is the moral-hazard effect and $PI = \hat{\delta}_x - \hat{\rho}/\hat{\pi}$ private information; $\hat{\theta}$ is the moral-hazard share of the gap and θ cap its one-sided 95% upper bound, obtained by inverting $w(\theta_0)$ in every column. *Main* is the specification of Table 6: a 5% threshold for a program counting as using unconditional offers, programs averaging at least 20 offers per year over 2010–2012, clustering on the application set, and standard errors from a 200-replication cluster bootstrap. The following columns report a variety of robustness checks, changing elements of our primary specification. (1) replaces the bootstrap with standard errors propagated linearly from the three clustered inputs under an assumption of independence. (2) raises the threshold for using unconditional offers from 5% to 10%. (3) restricts to 2010–2015. (4), (5) and (6) set the baseline offer requirement to a mean of at least 0, 10 and 30 offers per year rather than 20. (7) clusters on the least-selective university in the application set rather than on the set itself, which is the only column that speaks to correlation across sets that share a university. (8) drops observations after a set’s first year of exposure, and is equivalent to $ATT(g, g)$ from Callaway and Sant’Anna (2021). Columns (2)–(8) all assume independence, as in (1); the gap in standard errors between (1) and Main shows the implication of this assumption.

Table A.5: Treatment effects across the distribution of achieved attainment

Quantile	(1) Smoothing (−.01, .01)	(2) Smoothing (−.05, .05)	(3) Smoothing (−.1, .1)
0.1	−0.005 (0.019)	−0.005 (0.018)	−0.023 (0.020)
0.2	−0.008 (0.015)	−0.026* (0.015)	−0.029** (0.014)
0.3	−0.008 (0.014)	−0.003 (0.014)	−0.029** (0.013)
0.4	0.003 (0.015)	−0.006 (0.014)	−0.016 (0.014)
0.5	0.014 (0.014)	0.018 (0.014)	0.003 (0.014)
0.6	0.007 (0.016)	0.017 (0.016)	0.020 (0.016)
0.7	0.005 (0.016)	0.015 (0.017)	0.026 (0.016)
0.8	0.041** (0.018)	0.029* (0.017)	0.055*** (0.018)
0.9	0.039** (0.018)	0.027 (0.018)	0.018 (0.019)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column reports the coefficient on first exposure from a quantile regression of standardized achieved age-18 attainment on the same controls and fixed effects as [Table 6](#)'s $\hat{\rho}$ (year fixed effects, application-set $\times$ post-first-exposure fixed effects, academic and demographic controls), estimated separately at each decile 0.1–0.9, with standard errors in parentheses. Columns differ in the smoothing bandwidth of the smoothed quantile regression, given in each column header. $N = 368,874$.

Table A.6: Always-taker robustness of the decomposition

Assumed always-taker effect	0 SD	−0.05 SD	−0.10 SD	−0.15 SD
ATT	0.030 (0.045)	0.026 (0.045)	0.022 (0.045)	0.018 (0.045)
PI	−0.164 (0.046)	−0.160 (0.046)	−0.156 (0.046)	−0.152 (0.046)
$\hat{\theta}$	−0.224 (0.332)	−0.193 (0.332)	−0.162 (0.332)	−0.131 (0.332)
θ cap	0.324	0.355	0.386	0.417

Relaxes the assumption in [Section 3.1](#) that there are no always-takers, and recomputes bounds on the share of the conditional gap which is composed of moral hazard. [Table 5](#), column (2) puts this share at $\hat{p}_A = 0.0129$ (SE 0.00201), against a complier share of $\hat{\pi} = 0.143$; write $\xi_C = \hat{\pi}/(\hat{\pi} + \hat{p}_A) = 0.917$ and $\xi_A = \hat{p}_A/(\hat{\pi} + \hat{p}_A) = 0.083$ for the resulting weights on compliers and always-takers among the treated. Each column assumes always-takers have the stated moral-hazard effect (in place of the complier LATE of [Table 6](#)) and recomputes $ATT = \xi_A$ (assumed effect) + $\xi_C(\hat{\rho}/\hat{\pi})$, $PI = \hat{\delta}_X - ATT$, and $\theta = ATT/\hat{\delta}_X$. Rather than treat the assumed effect as known with certainty, we give it a standard error scaled for the always-takers' smaller population. Standard errors on ATT, PI and θ propagate this alongside the analytic clustered standard errors of $\hat{\pi}$, $\hat{\rho}$ and $\hat{\delta}_X$, under an assumption of independence between all four, extending the approach of column (1) of [Table A.4](#). θ cap is the one-sided 95% upper bound on θ , obtained the same way as in [Table A.4](#) by inverting $w(\theta_0)$ on the resulting blended estimate. $\hat{\pi}$, $\hat{\rho}$, $\hat{\delta}_X$, $\hat{w}$ and its t -statistic are unaffected by the assumed always-taker effect and are omitted; see [Table 6](#) or [Table A.4](#)'s Main column for their values. Sample includes 1,700,457 students for the gap, 366,288 for the exposure effects, and 97,730 application-set clusters.

Table A.7: Convergence diagnostics for the structural estimates

	ESS	$\hat{R}$	Geweke
<i>Preference stage — 50,000 draws</i>			
Constant	107	1.046	−2.13
Can live at home	2,331	1.002	1.64
× Low-SES	2,042	1.002	0.97
σ_η^2	384	1.006	−0.57
Same region	368	1.002	0.18
σ_v^2	118	1.007	0.53
Belief acceptance		0.630	
Converged		5 of 6	
<i>Signal stage — 250,000 draws</i>			
λ_i at mean ability	52,510	1.000	−0.17
Gradient, per SD	49,711	1.000	0.26
Female	54,743	1.000	0.66
σ_ε^2	5,781	1.000	−0.19
σ_ψ^2	73	1.026	4.11
Belief acceptance		0.938	
Converged		4 of 5	

Diagnostics for [Table 8](#). A parameter counts as converged when split- $\hat{R} \leq 1.01$ and the effective sample size is at least 20; the sampler separately flags $|Geweke| > 2$. In the preference stage only the constant fails, and it is weakly identified by construction. In the signal stage only σ_ψ^2 fails, and its posterior standard deviation in [Table 8](#) is likely understated.

B Data and Variable Construction

In this section, we describe in detail the definition and construction of variables.

Table B.1: Variable definitions

Variable	Definition
Unconditional offer	Indicator equal to 1 if an offer is marked as unconditional using UCAS flags, defined at the offer level.
Major	A specific JACS3.0 principal subject code; the Joint Academic Coding System (v3.0) was the standard subject-classification scheme in UK higher education prior to 2019.
Program	A particular combination of university and major. Programs may consist of multiple courses (the true unit of application); we aggregate offers and admission across all courses in a given program.
Application set	The set of programs applied to by a given student, defined at the student level.
Uses unconditional	Indicator equal to 1 if more than 5% of a program's offers in a given year are unconditional offers, defined at the year-program level.
Exposed to unconditional offers	Indicator equal to 1 if any program in a given application set uses unconditional offers, defined at the application set level.
First exposure	Indicator for whether an application set is exposed to unconditional offers for the first time, defined at the year-application set level.
Later exposure	Indicator for whether an application set is exposed to unconditional offers in a period after first exposure, defined at the year-application set level.
Take-up	Indicator for take-up of an unconditional offer, which occurs either through ranking a binding unconditional offer first, or ranking a standard unconditional offer first or second. Referred to as "Take-up" or "Uncond. take-up" in the exhibits. Defined at the student level.
Age-18 scores (z)	A student's aggregate achievement on their age-18 examinations (which may include both A-level and BTECs). If a student took more than three such exams, we consider the three exams that were most taken by other students in their primary major of application. Grades are converted to a numerical value using "tariff points": A* = 56, A = 48, B = 40, C = 32, D = 24, E = 16, U = 0. BTEC grades are converted using the same UCAS Tariff scale, applied to whichever qualification size a student holds (e.g., for a BTEC National Extended Diploma, sized to three A-levels: D*D*D* = 168, D*D*D = 160, D*DD = 152, DDD = 144, DDM = 128, DMM = 112, MMM = 96, MMP = 80, MPP = 64, PPP = 48; for a BTEC National Diploma, sized to two A-levels: D*D* = 112, DD = 96, DM = 80, MM = 64, MP = 48, PP = 32). We normalize scores to mean 0, standard deviation 1, within groups of students taking the same number of A-levels and the same number of BTEC exams.
Teacher-predicted age-18 scores (z)	A student's aggregate achievement, as predicted by their teachers, on their age-18 examinations. A small minority of students have a completed A-level exam at the time of their applications; for these students, we replace the relevant predicted grade with their achieved grade. Variable construction proceeds identically to Age-18 scores (z).

Continued on next page

Table B.1 – continued from previous page

Variable	Definition
Teacher-predicted age-18 scores (categorical)	A student's aggregate achievement, as predicted by their teachers, on their age-18 examinations. Identical to Teacher-predicted age-18 scores (z), but rather than standardizing the scores, we create an indicator variable for each unique combination of (score 1, test type 1, score 2, test type 2, score 3, test type 3).
Age-16 test scores (z)	Student's achievement on their age-16 examinations (GCSEs). We convert percentiles on the exam to a z score using the inverse normal CDF. Additionally includes a separate indicator for missing values.
Socioeconomic quintile	Quintile (1–5, 1 most deprived) of the Index of Multiple Deprivation (IMD) associated with a student's postcode.
Low-SES	Indicator for whether a student's postcode belongs to the bottom two quintiles of IMD.
Female	Indicator in UCAS data of whether a given student is female.
Independent school	Indicator for whether a student attends an independent school.
Expected achievement (z)	Fitted values from a regression of Age-18 scores (z) on Teacher-predicted age-18 scores (categorical), Age-16 test scores (including indicator for missing values), Low-SES, and Female using data from 2010–2013. Absent moral hazard, this is the best prediction available of student achievement given their prior observable characteristics.
Distance (d_{is})	Straight-line distance in kilometers between a student's postcode and the program's institution.
Commuting range	Indicator equal to 1 if $d_{is} < 40\text{km}$, i.e. close_{is} in Section 4.1.4 .
Region match (m_{is})	Indicator for whether a student's home region and a program's region are the same, using the 12 International Territorial Level 1 (ITL1) regions defined by the Office for National Statistics.
Matriculation	Indicator for whether a student enrolls at a university in a given year (e.g., at age 18, 19, ...).
Years enrolled	Cumulative number of years a student is observed enrolled in HESA by a given age (e.g., by age 24), used in Section 3.2 .
Degree completion / classification	Indicator for completing a degree by a given age, and the classification received (first-class honors, 2:1, 2:2, third-class honors), from HESA.

C A Model of Unconditional Offers

Through a theoretical model, we show that both moral hazard and private information can generate the taker–decliner achievement gap.

To frame these two forces, we present a stylized model of the unconditional-offer market in which both are present. Our model shows that unconditional offers can be made, and taken, alongside either force; that either force alone produces a gap between takers and decliners of the kind we observe, so the gap does not tell them apart; and that although adopters anticipate the gap, programs can still be hurt.

A unit mass of students is matched to four programs $k \in \{a, b, c, d\}$, each with capacity $\frac{1}{4}$. A student's ability is $A = v_1 + v_2$, with v_1 and v_2 independent and uniform on $[0, 1]$: v_1 is public (age-16 scores, teacher-predicted grades), while v_2 is unobserved by all parties until the age-18 examination. With probability p , independent of ability, a student privately observes v_2 before the market opens; otherwise she observes nothing. A student who takes an unconditional offer prepares less for the examination: her realized ability is $A - s$, with $s \geq 0$. Because her place is secured, the loss falls on the program that admitted her and on no one's placement.³¹ Finally, a share q of unconditional-offer holders decline for reasons unrelated to ability, such as a preferred offer elsewhere, and rank their conditional offers instead.³² Programs are sophisticated and aware of p and s , but not which students received a signal. Programs maximize the total realized ability of their intake.³³ A student who enrolls at k receives utility u_k , with $u_a > u_b > u_c > u_d$, and evaluates lotteries with a strictly increasing, strictly concave U .

Timing follows the UK admissions cycle. Students apply to all programs before v_2 is realized. Programs can make two types of offers: offers which are conditional on the revelation of v_2 , and unconditional, in which admission is guaranteed without regard to v_2 . Students can either rank their conditional offers, or commit to one unconditional offer and forfeit all other offers.

An equilibrium is a profile in which each program's set of unconditional offers maximizes its payoff, each student's take-up decision maximizes her expected utility, and conditional admission clears the market. Strategies are the set of students given unconditional offers by each program — in equilibrium, an interval of observable ability, the program's *unconditional-offer range*. The remaining conditional admission thresholds are set by market clearing. To simplify the analysis, we assume that students break any ties in favor of unconditional offers, and that programs make unconditional offers only when they are strictly profitable. We characterize the equilibrium of this market in [Proposition C.1](#). Four equilibria, one for each combination of the two forces, are shown in [Figure C.1](#). In each, one or both of the middle-tier programs give unconditional offers to a range of observably strong students, and the remainder of the class is admitted conditionally.

Proposition C.1 (Equilibrium unconditional-offer ranges). For every p and s the equilibrium is unique:

- (i) a and d never make unconditional offers. b and c each offer to an interval of observable ability, $[\underline{x}_k(p, s), \bar{x}_k]$; programs break even at the floor, and uninformed students stop accepting at the ceiling. The two intervals are disjoint, with c 's below b 's.
- (ii) Unconditional-offer ranges are widest at $p = s = 0$. Program k makes unconditional offers if and only if $s < \hat{s}_k(p)$, where the frontier $\hat{s}_k$ is strictly decreasing in p , positive at $p = 0$ whenever k offers at all, and reaches zero at some $p < 1$: enough private information ends unconditional offers on its own, as does a

³¹In this model we take no stand on the cost of shirking to the student herself, which matters to the extent that attainment affects later outcomes ([Section 3.2](#)); in the model, students value only their placement.

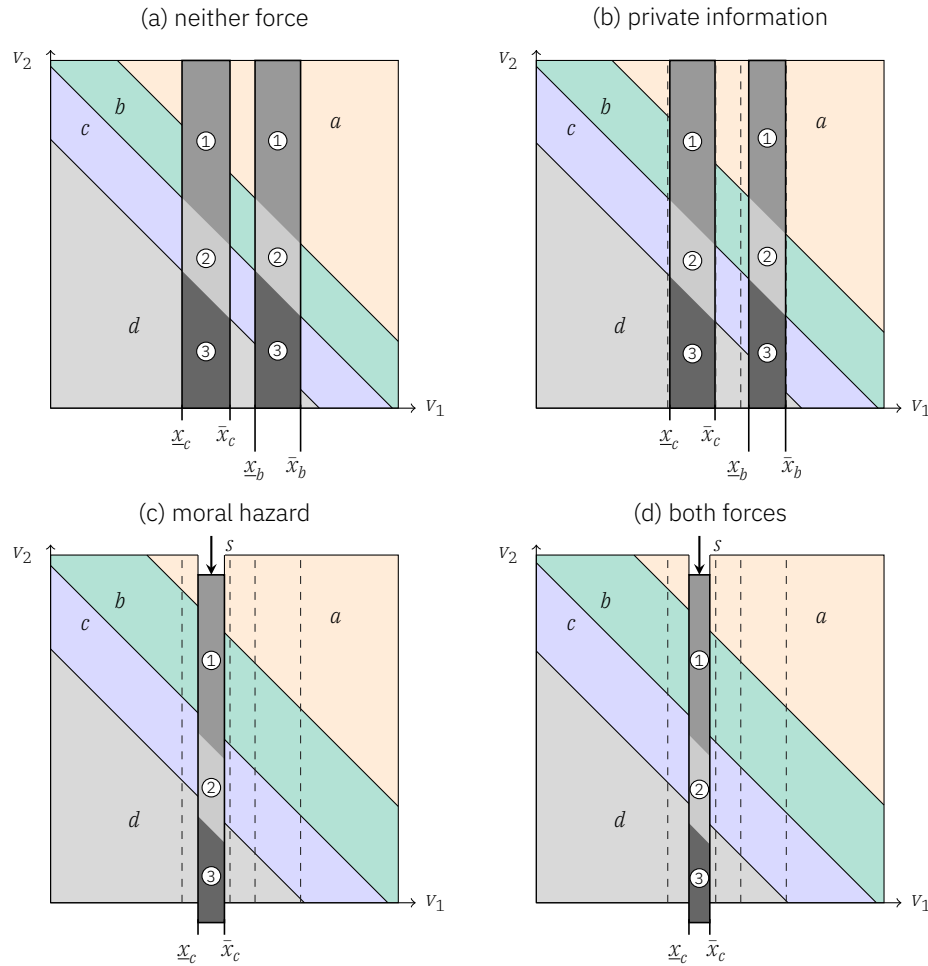
³² q multiplies the value of every unconditional offer and every taker mass by $1 - q$ and changes nothing else ([Appendix D](#)). We set $q = 0.2$ in [Figure C.1](#), so that every panel has decliners to compare takers with.

³³This is equivalent to maximizing average quality and enforcing capacity $1/4$. Because total enrolled ability is fixed when $s = 0$, the game among programs is then constant-sum, and every equilibrium is purely redistributive; when $s > 0$, total attainment falls by s times the mass of takers.

large enough attainment loss. Which program's frontier lies inside the other's depends on exact utility values (Appendix D.5); under the setup of Figure C.1, b 's lies inside c 's, so as either force grows both programs offer, then only c , then neither. Beyond both frontiers no program makes unconditional offers, and the assignment with conditional offers alone is the unique equilibrium.

Proof. See Appendix D.5. $\square$

Figure C.1: Equilibrium unconditional-offer ranges and take-up in the (v_1, v_2) plane, by force.



Notes: Axes are observable ability v_1 and the unobserved component v_2 ; ability $A = v_1 + v_2$ is constant along diagonals. Colored bands are the conditional-admission assignment (d, c, b, a from the bottom left), separated by the solid diagonals, the equilibrium thresholds $\underline{a}_k$. The vertical gray strips are the unconditional-offer ranges of c (left) and b (right), from floor $\underline{x}_k$ to ceiling $\bar{x}_k$; the dashed verticals in (b)–(d) mark the ranges of (a), for comparison. Within each strip, ① is the gain cell, takers who would otherwise have placed above k , each displacing a worse conditional admit; ② the neutral cell, takers who would have enrolled at k anyway; ③ the loss cell, takers who would have placed below k , each displacing a better conditional admit. Informed students, absent in (a) and (c), decline in ① and accept in ② and ③. The share q of holders who decline regardless, for reasons unrelated to ability, are spread uniformly over every cell and are not drawn, for ease of viewing. Under moral hazard, (c) and (d), takers score s less than they would have, so each strip is drawn shifted down by s , at the takers' realized v_2 . Panels use $u = (3, 2.5, 2, 1)$, $U = \ln$, $q = 0.2$, $p = 0.104$, and $s = 0.057$. In (b) and (c), this results in a taker–decliner achievement gap of $-0.139SD$ (Table 4). b makes no unconditional offers in (c) and (d). The four equilibria are detailed in Appendix D.3.

Unconditional offers grant insurance to students uncertain about their placement; this can benefit a program when it attracts students who would otherwise have enrolled above it, though unconditional offers do not always benefit adopting programs relative to an economy with only conditional offers. In every panel of [Figure C.1](#), c gains relative to the assignment with conditional offers alone and a and b lose. Program c 's range holds students who would otherwise have placed at b ; b 's unconditional offers attract students from a , but this gain is offset by the loss of students to c . Though b would prefer that unconditional offers were banned, it uses unconditional offers anyway because staying out would be worse: in panel (a), b 's payoff is 1.138 with conditional offers alone, 1.112 when it offers alongside c , and 1.088 if c uses unconditional offers but b does not. Private information and moral hazard both reduce the value of these offers to programs, but in different ways. Private information leads students with high v_2 to decline, reducing the expected value of an unconditional offer to the program. Moral hazard acts as a "tax" on unconditional offers, lowering the value of every unconditional offer by s times its take-up rate. Once the two forces together leave the offer at a program's ceiling unprofitable, that program stops offering altogether.

Both forces generate a gap between takers and decliners. Under private information, declining is itself the signal: the informed decliners are the students who know they would place above the program, so the gap is the distance between the strong students who leave and the average student who stays. This creates a "winner's curse" from the program's perspective: an informed student who takes an unconditional offer knows she is weak, while the uninformed take regardless. Under moral hazard, takers and decliners are the same students until the examination, and the gap is what the examination reveals. [Proposition D.6](#) gives the gap in closed form, as a private-information term that is nonzero if and only if $p > 0$ plus the tax s ; the same gap arises from a continuum of (p, s) pairs, and so the gap itself is not sufficient to distinguish the two. We show this through the parameters chosen for [Figure C.1](#): in (b), $(p, s) = (0.104, 0)$, in (c), $(p, s) = (0, 0.057)$; however, both produce the same gap of -0.139SD between takers and decliners, the conditional gap of [Table 4](#). Together, in panel (d), they produce -0.278SD .

In both cases, programs respond by shrinking their unconditional-offer ranges, but the two forces have different welfare consequences for programs and students, and thus policy implications. Private information decreases the benefits of unconditional offers, redistributing program welfare, but leaving its total constant. Ex-post student welfare (as measured after final matching) is constant under either force, since capacities fix the placements; ex-ante student welfare (as measured during the ranking phase) falls for uninformed students as either force shrinks the ranges and some lose the insurance of unconditional offers, while informed students are weakly better off for knowing v_2 . Moral hazard, by contrast, reduces the total welfare available to programs, in addition to redistributing it. Beyond the insurance channel, moral hazard impacts student welfare ambiguously, and depends on the sign of returns to age-18 exams, which the model does not price; returns encompass many components, such as human capital, non-pecuniary costs of studying for tests, and direct labor market effects. Under private information, a ban on unconditional offers is welfare-reducing for students and redistributive among programs, favoring those that lose from the offers, a and b ; under moral hazard, a ban is welfare-improving for programs in total, with ambiguous impacts on students.

D Appendix for the Theoretical Model

D.1 Timing

- | | | |
|---------|--------|--|
| $t = 0$ | apply | Students submit applications to all four programs. |
| $t = 1$ | offer | Programs make offers: conditional (admitting if the realized A meets the program's threshold) or unconditional (made before the examination, and so conditioned on v_1 alone). |
| $t = 2$ | rank | Students privately learn v_2 with probability p , then rank up to four offers. Students can either rank all conditional offers, or rank a single unconditional offer and forfeit all remaining offers. A share q of unconditional-offer holders, drawn independently of ability and information, rank their conditional offers regardless. |
| $t = 3$ | accept | Students who ranked conditional offers take a test, revealing $A = v_1 + v_2$. Students are accepted to the best-ranked program for which they meet the conditions. Students who took an unconditional offer score $A - s$, and keep their place. |

Equilibrium. Students best respond by [Lemma D.2](#). Because unconditional-offer ranges cover a nonnegligible share of capacity, takers shift the admission thresholds, so an equilibrium is a joint fixed point of unconditional-offer ranges and market clearing.³⁴ [Proposition C.1](#) characterizes the unconditional-offer sets, [Appendix D.3](#) computes the fixed point, and [Lemma D.5](#) shows it is unique.

D.2 Lemmas

We first give an overview of this section. [Appendix D.2](#) proves three lemmas about a single unconditional offer. [Lemma D.1](#) derives the assignment with conditional offers only, the benchmark. [Lemma D.2](#) characterizes who accepts an unconditional offer: uninformed students below a ceiling in v_1 , and informed students who would have placed at or below the program. [Lemma D.3](#) values a marginal unconditional offer: each taker is worth her ability net of the edge of the program's band that she crosses, and net of the s she forgoes by shirking; informed students accept exactly when the first part is nonpositive. [Appendix D.3](#) then treats unconditional-offer ranges of positive mass, which move the thresholds: [Lemma D.4](#) shows that optimal unconditional-offer sets are intervals, so that an equilibrium is a solution of a small system of equations, and [Lemma D.5](#) shows that the solution is unique. [Appendix D.5](#) proves [Propositions C.1](#) and [D.6](#) from these ingredients.

Define the *target* of an unconditional offer as the observable ability $x = v_1$ of the student who receives it. A program's *ceiling* $\bar{x}_k$ is the highest target at which uninformed students accept its offer ([Lemma D.2](#)), and its *floor* $\underline{x}_k(p, s)$ is the lowest target at which the offer breaks even ([Lemma D.4](#)); we write $\underline{x}_k(p)$ for the floor at $s = 0$ and $\underline{x}_k$ for the floor at $p = s = 0$. Primed objects (thresholds α'_k , ceilings $\bar{x}'_k$, floors $\underline{x}'_k(p)$, exits $\hat{p}'_k$) are evaluated at the conditional-only benchmark of [Lemma D.1](#), where [Lemmas D.2](#) and [D.3](#) analyze a single offer; unprimed objects are equilibrium values.

³⁴Conditional admission is not strategic: thresholds clear the market. Strategic conditional admission would not help either. Committing to reject an available student only passes her down the ranking and fills the seat from below, since the clearing thresholds above do not depend on whom a program rejects beneath them.

Our lemmas hold beyond the baseline economy (equal capacities, hence quartile thresholds, and $u = (3, 2.5, 2, 1)$ with $U = \ln$), which we use only for numerical evaluations. The proofs rely on three conditions, flagged where used and all satisfied at the baseline economy.

- (C1) Every band is narrower than the support of v_2 , so that students who do not observe v_2 have uncertainty about their program assignment.
- (C2) At the targets considered, $\underline{\alpha}'_k - x$ and $\bar{\alpha}'_k - x$ lie in $(0, 1)$, so that an offeree at target x may place below, at, or above k depending on her v_2 .
- (C3) Unconditional offers are viable for some k , i.e., $\bar{x}'_k > \underline{x}'_k$, so that some target is both profitable for k and accepted by uninformed students.

Preferences enter only through the ceilings $\bar{x}'_k$; the floors $\underline{x}'_k(p)$ are pure geometry. The one claim that depends on the baseline is the exit order in [Proposition C.1](#), which reverses when u_b is close enough to u_a ([Appendix D.5](#)). The two frictions enter in two places only. Moral hazard s is a per-taker cost in [Lemma D.3](#) and nowhere else: a taker's place is secured, so her lower score moves no threshold. The random-decline share q multiplies every taker mass, in [Lemma D.3](#) and in [Equation \(19\)](#), by $1 - q$, and nothing else: a random decliner is an ordinary conditional applicant. Every sign below and every marginal quantity is therefore invariant to q ; at positive scale q enters only by shrinking the offers ($\hat{p}_b$ falls from 0.575 to 0.557 at $q = 0.2$), and the tables report $q = 0$ unless stated.

Lemma D.1 (Benchmark assignment). With conditional offers only, the assignment that clears the market is unique and assortative: program k enrolls the students with $A \in [\underline{\alpha}'_k, \bar{\alpha}'_k)$, where the admission thresholds $(\underline{\alpha}'_a, \underline{\alpha}'_b, \underline{\alpha}'_c, \underline{\alpha}'_d) = (2 - \sqrt{\frac{1}{2}}, 1, \sqrt{\frac{1}{2}}, 0)$ are the quartiles of the distribution of A , and $\bar{\alpha}'_k$ is the threshold of the program ranked directly above k : $\bar{\alpha}'_c = \underline{\alpha}'_b$, $\bar{\alpha}'_b = \underline{\alpha}'_a$, and $\bar{\alpha}'_a = 2$, the top of the support. The assignment is the same for every p .

Proof. A has distribution $F(\alpha) = \frac{\alpha^2}{2}$ for $\alpha \leq 1$ and $1 - \frac{(2-\alpha)^2}{2}$ for $\alpha \geq 1$, strictly increasing on $[0, 2]$. Students share the ranking $u_a > u_b > u_c > u_d$, so in any market-clearing outcome in which each student enrolls at the highest-ranked program that admits her, thresholds are ordered and each intake is the interval between consecutive thresholds. Capacities of $\frac{1}{4}$ then force $1 - F(\underline{\alpha}'_a) = \frac{1}{4}$, $1 - F(\underline{\alpha}'_b) = \frac{1}{2}$, $1 - F(\underline{\alpha}'_c) = \frac{3}{4}$, and $\underline{\alpha}'_d = 0$; strict monotonicity of F makes the solution unique. The argument uses only the common ranking, atomlessness of F , and capacities summing to one; with general capacities, the thresholds are the corresponding quantiles. Nothing is chosen before the examination, so nothing depends on p . $\square$

Write $\mu(A)$ for the program assigned to ability A under [Lemma D.1](#). Program k 's band is $[\underline{\alpha}'_k, \bar{\alpha}'_k)$: its own threshold is the lower edge, and the threshold of the program directly above is the upper edge, so an overline on α is the underline of the next program up. Define the *net contribution* of a student of ability A who enrolls at k by taking an unconditional offer,

$$\varphi_k(A) \equiv \max\{A - \bar{\alpha}'_k, 0\} - \max\{\underline{\alpha}'_k - A, 0\},$$

her distance outside k 's band, signed: positive if she would have placed above k , zero if at k , negative if below. A taker who would have placed below k costs k its worst student, of ability $\underline{\alpha}'_k$. A taker who would have placed above k costs k its *best* student: she leaves a vacancy above, which is refilled from the top of k 's band, so k keeps her at the price of a student of ability $\bar{\alpha}'_k$. [Lemma D.3](#) makes this exact.

Lemma D.2 (Acceptance). Consider a single unconditional offer from $k \in \{b, c\}$ to a student with $v_1 = x$. One offer has measure zero, so thresholds stay at the benchmark values of [Lemma D.1](#); primes mark objects evaluated at those thresholds.

- (i) An uninformed holder accepts if and only if $x < \bar{x}'_k$. The *ceiling* $\bar{x}'_k$ is the unique solution in $(0, \underline{\alpha}'_k)$ of $W_k(x) = U(u_k)$, where $W_k(x) \equiv E[U(u_{\mu(A)}) \mid v_1 = x]$ is the *lottery utility* of keeping the conditional offers: continuous, piecewise linear, and strictly increasing in x .³⁵ At the baseline, $\bar{x}'_c = 0.495$ and $\bar{x}'_b = 0.698$.
- (ii) An informed holder accepts if and only if she would have placed at or below k , that is, below the upper edge of k 's band, $A < \bar{\alpha}'_k$, for every increasing U : equivalently, if and only if $\varphi_k(A) \leq 0$.

Proof. Conditional on $v_1 = x$, $A \sim U[x, x + 1]$, so the lottery utility is

$$W_k(x) = \int_0^1 U(u_{\mu(x+t)}) dt,$$

continuous and piecewise linear, with derivative $U(u_{\mu(x+1)}) - U(u_{\mu(x)})$ wherever x and $x + 1$ are interior to bands. Raising the target shifts probability from the worst reachable band to the best, so the derivative is positive as long as $\mu(x + 1)$ is ranked strictly above $\mu(x)$, which (C1) guarantees. At $x = 0$ the support $[0, 1]$ of A meets only d 's and c 's bands, so $W_c(0) = \underline{\alpha}'_c U(u_d) + (1 - \underline{\alpha}'_c) U(u_c) < U(u_c)$;³⁶ at $x = \underline{\alpha}'_c$ the support $[\underline{\alpha}'_c, \underline{\alpha}'_c + 1]$ lies weakly above $\underline{\alpha}'_c$ and strictly above $\underline{\alpha}'_b$ with positive probability, so $W_c(\underline{\alpha}'_c) > U(u_c)$. Hence W_c crosses $U(u_c)$ exactly once on $(0, \underline{\alpha}'_c)$, at $\bar{x}'_c = 0.495$ for the baseline; the indifferent student at the ceiling accepts by convention. The same argument with u_b gives $\bar{x}'_b = 0.698$. For (ii), an informed holder knows A and compares the certain u_k to the certain $u_{\mu(A)}$: she accepts if and only if $\mu(A)$ is k or worse, i.e. $A < \bar{\alpha}'_k$, for any increasing U , and at $\mu(A) = k$ she accepts by convention. Since $\varphi_k(A) \leq 0$ exactly when $A < \bar{\alpha}'_k$ (up to the measure-zero boundary), this is the last claim. $\square$

Remark (acceptance at other thresholds). Because each student is atomistic, [Lemma D.2](#) applies verbatim at any prevailing thresholds, including when she is one of a positive mass of offer holders: acceptance is a cutoff in x , with W_k , μ , φ_k , and the ceiling evaluated at those thresholds, and an informed holder accepts if and only if $\varphi_k(A) \leq 0$ there. This requires only (C1) at those thresholds, which holds at every equilibrium in [Appendix D.3](#).

³⁵The ceiling carries a prime because the lottery depends on the thresholds through μ ; at the equilibrium thresholds of [Appendix D.3](#) it is a different number, $\bar{x}_k$.

³⁶In general, $W_k(0) < U(u_k)$ is the condition that some uninformed student accepts, which is part of viability (C3).

For an offer at target x , the offeree's v_2 falls into one of three cells: the *loss cell* $v_2 < \underline{\alpha}'_k - x$, of length $\ell_k(x) \equiv \underline{\alpha}'_k - x$, where she would have placed below k ; the *neutral cell* $\underline{\alpha}'_k - x \leq v_2 < \bar{\alpha}'_k - x$, where she would have placed at k ; and the *gain cell* $v_2 \geq \bar{\alpha}'_k - x$, of length $g_k(x) \equiv x + 1 - \bar{\alpha}'_k$, where she would have placed above k . Under (C2) all three cells are nonempty.

Lemma D.3 (Value of a marginal unconditional offer). Let program k make unconditional offers to every student with $v_1 \in [x, x + \varepsilon)$, with students following Lemma D.2. The prevailing thresholds may be the benchmark thresholds of Lemma D.1 or those induced by other programs' unconditional offers (Appendix D.3); φ_k and the cells are evaluated at them. The change in k 's payoff is $\varepsilon V_k(x) + O(\varepsilon^2)$, where

$$V_k(x) = (1 - q) \left[(1 - p) \mathbf{1}\{x < \bar{x}_k\} \int_0^1 \varphi_k(x + v_2) dv_2 + p \int_0^1 \min\{\varphi_k(x + v_2), 0\} dv_2 - s \tau_k(x) \right],$$

and $\tau_k(x) = (1 - p) \mathbf{1}\{x < \bar{x}_k\} + p \int_0^1 \mathbf{1}\{\varphi_k(x + v_2) \leq 0\} dv_2$ is take-up at x among holders not subject to the random decline: an uninformed taker contributes her full net contribution, an informed taker, who accepts exactly when $\varphi_k \leq 0$, contributes only its negative part, and every taker costs s .

Proof. Capacities bind and clearing is assortative, so when a taker of ability A enters k 's intake exactly one student leaves it, and she is always at a threshold. If the taker would have placed above k , the vacancy she leaves is filled from the top down, each program taking the best student of the program below, until the program directly above k takes k 's best student, of ability $\bar{\alpha}_k$. If she would have placed below k , k sheds its worst student, of ability $\underline{\alpha}_k$, and the surplus passes down in the same way to the program she vacated. If she would have placed at k , she replaces herself. In each case k 's net is $\varphi_k(A)$, and no other program's intake enters k 's payoff. The taker then scores $A - s$ rather than A ; since her place is secured, the shortfall changes no assignment and is a cost of s to k alone. Summing over the strip $[x, x + \varepsilon)$: by Lemma D.2, the uninformed accept if and only if $x < \bar{x}_k$ and are uniform in v_2 , while the informed accept if and only if $\varphi_k(A) \leq 0$; the share q who decline regardless are uniform in v_2 and independent of information, so they scale every term by $1 - q$ and enter the conditional pool as ordinary applicants. The takers have mass $O(\varepsilon)$, so thresholds move by $O(\varepsilon)$ and each displaced student has ability equal to the threshold up to $O(\varepsilon)$; offerees whose acceptance flips because thresholds moved lie within $O(\varepsilon)$ of the ceiling $\bar{x}_k$ or of the boundary $\varphi_k = 0$, a set of measure $O(\varepsilon^2)$ within the strip. $\square$

Lemma D.3 gives the value of a target in closed form. With the cell lengths ℓ_k and g_k both in $(0, 1)$ (C2), $\int_0^1 \varphi_k = \frac{1}{2} (g_k^2 - \ell_k^2)$, $\int_0^1 \min\{\varphi_k, 0\} = -\frac{1}{2} \ell_k^2$, and the informed accept on the loss and neutral cells, of combined length $1 - g_k$, so $\tau_k = 1 - p g_k$. Below the ceiling $\bar{x}_k$, where uninformed students

accept,

$$\frac{V_k(x)}{1-q} = \frac{1}{2} [(1-p)g_k(x)^2 - \ell_k(x)^2] - s(1-pg_k(x)) = G_k(x) - p\frac{g_k(x)^2}{2} - s(1-pg_k(x)), \quad (18)$$

where $G_k(x) \equiv \frac{1}{2} [g_k(x)^2 - \ell_k(x)^2]$ is the value with neither force: private information removes a fraction p of the gain cell and nothing else, and moral hazard charges s per taker on the take-up rate $1-pg_k$. Above the ceiling only informed students accept, so $V_k = -(1-q)p[\frac{1}{2}\ell_k^2 + s(1-g_k)] \leq 0$. Raising the target by dx lengthens the gain cell and shortens the loss cell by dx each, so below the ceiling V_k is strictly increasing in x , with derivative $(1-q)[(1-p)g_k + \ell_k + sp] > 0$; it is strictly decreasing in s , and strictly decreasing in p wherever $g_k > 2s$, since an informed gain-cell student who declines costs k her average net contribution $g_k/2$ and saves it the s she would have forgone.

Remark (a and d). For a , $\bar{\alpha}'_a = 2$ is the top of the support, so the gain cell is empty: every taker has $\varphi_a \leq 0$, strictly below a 's band, and $V_a < 0$ at every target and every p . For d , $x+1 > \underline{\alpha}'_c$ for every x by (C1), so an uninformed holder of d 's offer is assigned above d with positive probability if she keeps her conditional offers, and never accepts; an informed holder accepts only if she would have placed at d anyway, a wash. Neither program has a strictly profitable target at $s = 0$, and $s > 0$ only lowers every value, so by convention neither makes unconditional offers.

D.3 Positive scale

Lemmas D.2 and D.3 value a single unconditional offer at fixed thresholds, which is enough to sign every claim in **Propositions C.1 and D.6 (Appendix D.5)**, because an unconditional-offer set of mass ε moves every threshold by $O(\varepsilon)$. In equilibrium, an unconditional-offer range of positive mass moves the thresholds it is evaluated against. In this section, we find said equilibrium; **Lemma D.4** reduces each adopter's strategy to one number, and **Equation (19)** then gives the thresholds.

Lemma D.4 (Interval unconditional-offer sets). Fix any thresholds satisfying (C1) and any p and s . For an adopter k , the value $V_k(x)$ of **Equation (18)** is strictly increasing in x below the ceiling $\bar{x}_k$ and nonpositive above it. Every optimal unconditional-offer set is therefore an interval $[\underline{x}_k, \bar{x}_k)$ that ends at the ceiling and is determined by its floor $\underline{x}_k$, and the derivative of the adopter's payoff in its floor is the value of the floor target, so an optimal floor breaks even: $V_k(\underline{x}_k) = 0$ at the thresholds the range induces.

Proof. Below the ceiling, $V'_k = (1-q)[(1-p)g_k + \ell_k + sp] > 0$. Above it only informed students accept, each has $\varphi_k \leq 0$ by **Lemma D.2(ii)**, and each costs s . Since **Lemma D.3** holds at any thresholds that clear the market, including those induced by the adopter's own unconditional-offer set, shifting a small mass of offers from a target x to a target $x' > x$ below the ceiling changes the adopter's payoff by $V_k(x') - V_k(x) > 0$ times that mass, to first order. An optimal set therefore contains every target above any target it contains, up to the ceiling. Removing a marginal strip at the floor is **Lemma D.3** with a negative ε , so the derivative of the payoff in the floor is $V_k(\underline{x}_k)$, which an optimal floor sets to zero. $\square$

We now examine market clearing conditions. Let $K \subseteq \{b, c\}$ be the set of adopters, each $k \in K$ offering on the strip $[\underline{x}_k, \bar{x}_k]$ of length L_k and midpoint m_k ; every offeree in the strip not subject to the random decline takes if uninformed, and takes if informed exactly when $A < \bar{\alpha}_k$ (Lemma D.2); the share q who decline regardless rank their conditional offers. Write $\bar{F}(\theta) \equiv 1 - F(\theta)$ for the mass of students with $A \geq \theta$ and κ_j for capacities. Students with $A \geq \underline{\alpha}_j$ either take an unconditional offer or enroll conditionally at j or above, so for each $j \in \{a, b, c\}$,

$$\bar{F}(\underline{\alpha}_j) = \sum_{i \geq j} \kappa_i + (1 - q)(1 - p) \sum_{i \in K, i < j} L_i (1 - \underline{\alpha}_j + m_i) - (1 - q) \sum_{i \in K, i \geq j} L_i (\underline{\alpha}_j - m_i). \quad (19)$$

Equation (19) is a seat count, and s does not appear in it: a taker's lower score moves no threshold, because her place is already hers. The seats at j and above, $\sum_{i \geq j} \kappa_i$, are filled either conditionally, by students with $A \geq \underline{\alpha}_j$ who took no unconditional offer, or unconditionally, by the takers of adopters at or above j . The mass $\bar{F}(\underline{\alpha}_j)$ therefore equals the seats, plus the takers with $A \geq \underline{\alpha}_j$ at adopters below j (counted in $\bar{F}$, but filling no seat at j or above), minus the takers with $A < \underline{\alpha}_j$ at adopters at or above j (filling a seat, but not counted in $\bar{F}$). In words, the threshold of j falls by the takers that adopters below j capture from above it (uninformed only, since informed takers of i have $A < \bar{\alpha}_i \leq \underline{\alpha}_j$), and rises by the takers that adopters at or above j admit from below it (informed and uninformed alike, since $A < \underline{\alpha}_j \leq \bar{\alpha}_i$). Both terms are linear in the strip endpoints because every cell boundary is interior (C2), which holds at every solution below; the two strips are also disjoint there, so no student holds two unconditional offers. An equilibrium with adopter set K is a solution of Equation (19) for $j = a, b, c$ together with, for each $k \in K$, the acceptance condition $W_k(\bar{x}_k) = U(u_k)$ and the floor condition $V_k(\underline{x}_k) = 0$, both evaluated at the equilibrium thresholds: $3 + 2|K|$ equations in as many unknowns.

Given the strip endpoints, Equation (19) is a quadratic in $\underline{\alpha}_j$ with one root in $[0, 2]$; the acceptance condition is linear in the ceiling, a weighted average of the thresholds (displayed in the proof of Lemma D.5); and by Equation (18) the floor condition $V_k(\underline{x}_k) = 0$ is, at $s = 0$, $\sqrt{1 - p} g_k = \ell_k$, the gain cell equal to $1/\sqrt{1 - p}$ times the loss cell, which is linear in the floor: $\underline{x}_k = [\underline{\alpha}_k - \sqrt{1 - p} (1 - \bar{\alpha}_k)] / (1 + \sqrt{1 - p})$. With $s > 0$ the floor condition is the quadratic $(1 - p) g_k^2 - \ell_k^2 = 2s(1 - p g_k)$, whose left side is increasing in the floor and right side decreasing, so it has one admissible root, increasing in s . The equilibrium is therefore a fixed point in the $2|K|$ endpoints. Solving, at $q = 0$, along p with $s = 0$ and along s with $p = 0$:³⁷

³⁷ As $L_k \rightarrow 0$ the thresholds return to the benchmark, so c 's unconditional-offer range vanishes exactly where the benchmark floor reaches the benchmark ceiling, $\hat{p}_c = 0.816$ (Appendix D.5). Adopter k fills $M_k = (1 - p)L_k + p \int_{\underline{x}_k}^{\bar{x}_k} (\bar{\alpha}_k - t) dt$ of its seats with unconditional admits and the rest conditionally at $\underline{\alpha}_k$; the last columns report M_k as a share of capacity.

p	s	$\underline{\alpha}_a$	$\underline{\alpha}_b$	$\underline{\alpha}_c$	$\underline{x}_b$	$\bar{x}_b$	$\underline{x}_c$	$\bar{x}_c$	M_b	M_c
0	0	1.136	0.997	0.796	0.566	0.727	0.397	0.524	64%	51%
0.2	0	1.212	0.987	0.770	0.621	0.722	0.400	0.519	37%	43%
0.35	0	1.250	0.986	0.754	0.657	0.717	0.411	0.514	20%	34%
0.5	0	1.276	0.986	0.739	0.692	0.712	0.427	0.509	6%	25%
0.6	0	1.286	0.988	0.730	—	—	0.442	0.505	—	18%
0.7	0	1.290	0.994	0.720	—	—	0.463	0.501	—	10%
0.8	0	1.293	0.999	0.709	—	—	0.490	0.496	—	1%
0	0.02	1.185	0.981	0.770	0.608	0.716	0.401	0.513	43%	45%
0	0.04	1.231	0.969	0.750	0.654	0.708	0.411	0.505	22%	38%
0	0.05	1.252	0.965	0.742	0.678	0.706	0.418	0.503	11%	34%
0	0.07	1.277	0.971	0.728	—	—	0.442	0.499	—	23%
0	0.09	1.287	0.990	0.713	—	—	0.476	0.496	—	8%

We now can determine threshold directions. The signs of the two sums in [Equation \(19\)](#) order the thresholds against the benchmark. No adopter lies below c , so only the negative sum enters c 's condition and $\underline{\alpha}_c > \underline{\alpha}'_c$; no adopter lies at or above a , so only the positive sum enters a 's and $\underline{\alpha}_a < \underline{\alpha}'_a$. For b , only the positive sum enters when c is the sole adopter, so $\underline{\alpha}_b < \underline{\alpha}'_b$; when b also adopts, its own takers below $\underline{\alpha}_b$ enter with the opposite sign, and the computed fixed point settles the comparison, $\underline{\alpha}_b < \underline{\alpha}'_b$ in every row of the table. Beyond $\hat{p}_c$ the system has no positive-length solution, so making no offers is an equilibrium against deviations of any scale.

The box collects the four equilibria of [Figure C.1](#): the unconditional-offer ranges, and everything market clearing implies from them. All four use $q = 0.2$, so that every panel has decliners; $p = 0.104$ and $s = 0.057$ are chosen so that each force alone reproduces the conditional taker–decliner achievement gap of -0.139SD ([Table 4](#)), with one standard deviation of A equal to $1/\sqrt{6}$.

The equilibria of Figure C.1.

1. The equilibrium is defined by unconditional-offer ranges. a and d make no unconditional offers; every uninformed offer holder accepts, an informed one accepts only if she knows she would place at or below the adopter, and a share q decline regardless.

	(a) neither $p = 0, s = 0$	(b) private information $p = 0.104, s = 0$	(c) moral hazard $p = 0, s = 0.057$	(d) both $p = 0.104, s = 0.057$
c offers unconditionally to $v_1 \in$	[0.378, 0.516)	[0.384, 0.514)	[0.424, 0.500)	[0.439, 0.499)
b offers unconditionally to $v_1 \in$	[0.588, 0.719)	[0.611, 0.717)	—	—

2. Through market clearing, we derive conditional thresholds $\underline{\alpha}_k$; these yield take-up among offer holders, the taker–decliner achievement gap, and takers' shortfall relative to the observable projection in c 's range (Proposition D.6; b 's in (b) are 0.763, -0.139 SD, 0.033SD); the share of each adopter's intake arriving unconditionally; and mean realized intake ability relative to the conditional-only benchmark (1.528, 1.138, 0.862, 0.472).

	(a) neither	(b) private information	(c) moral hazard	(d) both
Threshold $\underline{\alpha}_a$	1.192	1.218	1.277	1.281
$\underline{\alpha}_b$	0.983	0.983	0.970	0.978
$\underline{\alpha}_c$	0.773	0.763	0.730	0.724
Take-up in c 's range	0.800	0.761	0.800	0.759
Taker–decliner achievement gap (SD)	0	-0.139	-0.139	-0.278
Takers' shortfall vs. projection (SD)	0	0.033	0.139	0.173
Unconditional share of intake, b/c	42% / 44%	32% / 40%	— / 24%	— / 18%
Mean intake ability vs. benchmark, a	-0.045	-0.032	-0.004	-0.003
b	-0.026	-0.025	-0.024	-0.017
c	$+0.050$	$+0.040$	$+0.006$	$+0.004$
d	$+0.021$	$+0.018$	$+0.008$	$+0.006$
Total attainment vs. benchmark	0	0	-0.003	-0.003

Every adopter's intake exceeds what it would be if it alone stopped offering, the equilibrium condition. Relative to the benchmark, c and d gain while a and b lose in every panel; b offers in (a) and (b) because c does, and withdraws in (c) and (d) because $s = 0.057$ exceeds its frontier $\hat{s}_b(p)$. Total attainment is fixed at $s = 0$ and falls by s times the mass of takers otherwise.

b 's unconditional-offer range does not vanish at $\hat{p}'_b = 0.443$, where its benchmark floor reaches its benchmark ceiling (Appendix D.5) and where it still supplies 10% of b 's intake, but at $\hat{p}_b = 0.575$, the root of $V_b(\bar{x}_b) = 0$ at the lone-adopter equilibrium thresholds: b exits into a market where c still makes unconditional offers and depresses every rival's threshold, $\underline{\alpha}_a$ most of all, which enlarges b 's gain cell. For c the two exits coincide, $\hat{p}_c = \hat{p}'_c$, because no unconditional-offer range remains at c 's exit, so the last offer faces the benchmark thresholds. The exits in s behave the same way. At $p = 0$, c 's range vanishes at $\hat{s}_c = \hat{s}'_c = 0.100$, the value of its benchmark ceiling target per taker, $G_c(\bar{x}'_c)$; b 's vanishes at $\hat{s}_b = 0.060$ at scale against $\hat{s}'_b = 0.036$ at the margin, again because c 's remaining offers enlarge b 's gain cell. The frontier $\hat{s}_k(p)$ falls in p : at the margin it is $\hat{s}'_k(p) = [(1-p)g_k^2 - \ell_k^2] / [2(1-p)g_k]$ at the ceiling, and at scale $(\hat{s}_b, \hat{s}_c) = (0.051, 0.092)$ at $p = 0.104$ and $(0.042, 0.084)$ at $p = 0.2$.

Lemma D.5 (Uniqueness at scale). Fix $p < \hat{p}_c$ and write f for the density of A . Suppose every equilibrium floor is positive and, at each threshold, the density exceeds the

total uninformed offer mass entering that threshold's clearing condition ($f(\underline{\alpha}_k) > (1-p)L_c$, and $f(\underline{\alpha}_c) > L_c$).

- (i) For any floors, the clearing-and-acceptance subsystem has a unique solution.
- (ii) Write D for the total derivative in an adopter's own floor, with thresholds and ceilings adjusting by (i). Each adopter's floor condition is strictly increasing in its floor whenever the threshold bounding its gain cell responds less than one-for-one, $D\bar{\alpha}_k < 1$; the threshold bounding its loss cell never rises with the floor. Cross-effects between adopters have opposite signs.
- (iii) Under (ii), the equilibrium for each set of adopters is unique. At the baseline, (ii) holds with room to spare across the admissible floors: $D\bar{\alpha}_b \leq 0.29$ against the bound of one, and the floor conditions' own-derivatives are at least 0.39 (lone adopter) and 0.22 (two adopters).

Proof. (i) *The ceiling is a weighted average of the thresholds.* On the relevant branch, $W_c(x) = U(u_d)(\underline{\alpha}_c - x) + U(u_c)(\underline{\alpha}_b - \underline{\alpha}_c) + U(u_b)(\underline{\alpha}_a - \underline{\alpha}_b) + U(u_a)(x + 1 - \underline{\alpha}_a)$, so $W_c(\bar{x}_c) = U(u_c)$ solves to

$$\bar{x}_c = \omega_c \underline{\alpha}_c + \omega_b(\underline{\alpha}_b - 1) + \omega_a(\underline{\alpha}_a - 1),$$

$$\omega_c = \frac{U(u_c) - U(u_d)}{U(u_a) - U(u_d)}, \quad \omega_b = \frac{U(u_b) - U(u_c)}{U(u_a) - U(u_d)}, \quad \omega_a = \frac{U(u_a) - U(u_b)}{U(u_a) - U(u_d)},$$

positive weights summing to one, and $\bar{x}_b = \bar{x}_c + \omega_b$. Given floors and ceilings, thresholds clear uniquely (Lemma D.1 on the retaining population). Differentiating the clearing conditions in a ceiling,

$$\frac{\partial \underline{\alpha}_c}{\partial \bar{x}_c} = \frac{\underline{\alpha}_c - \bar{x}_c}{f(\underline{\alpha}_c) - L_c} \in (0, 1),$$

$$\frac{\partial \underline{\alpha}_b}{\partial \bar{x}_c} = -\frac{(1-p)(1 - \underline{\alpha}_b + \bar{x}_c)}{f(\underline{\alpha}_b) - (1-p)L_c} \leq 0, \quad \frac{\partial \underline{\alpha}_a}{\partial \bar{x}_c} = -\frac{(1-p)(1 - \underline{\alpha}_a + \bar{x}_c)}{f(\underline{\alpha}_a) - (1-p)L_c} \leq 0,$$

the first in $(0, 1)$ because $\bar{x}_c > L_c$ is exactly $\underline{x}_c > 0$. The map $\bar{x}_c \mapsto \sum_k \omega_k \underline{\alpha}_k(\bar{x}_c)$ therefore has slope $\omega_c \partial \underline{\alpha}_c / \partial \bar{x}_c$ plus nonpositive terms, strictly below one, so $\bar{x}_c - \sum_k \omega_k \underline{\alpha}_k(\bar{x}_c)$ is strictly increasing and vanishes exactly once. With two adopters the same partials bound the Jacobian of $(\bar{x}_b, \bar{x}_c) \mapsto$ ceilings; the map is a sup-norm contraction whenever each row's absolute sum is below one, which the density condition delivers (at the baseline the row sums never exceed 0.17).

(ii) *The floor condition is single-crossing.* Write $\Phi_c(\underline{x}_c) \equiv V_c(\underline{x}_c; \underline{\alpha}(\underline{x}_c))$ for the value of the floor target at the thresholds it induces, with $g = \underline{x}_c + 1 - \underline{\alpha}_b$ and $\ell = \underline{\alpha}_c - \underline{x}_c$, so $\Phi_c = \frac{1}{2}[(1-p)g^2 - \ell^2]$ and

$$\Phi'_c = (1-p)g(1 - D\underline{\alpha}_b) + \ell(1 - D\underline{\alpha}_c).$$

Differentiating the clearing and acceptance conditions in the floor and substituting

the acceptance identity $D\bar{x}_c = \sum_k \omega_k D\alpha_k$ gives

$$D\bar{x}_c - 1 = -\frac{L_c \Omega}{B} < 0,$$

$$\Omega \equiv \frac{\omega_c}{f(\alpha_c) - L_c} + (1-p) \left(\frac{\omega_b}{f(\alpha_b) - (1-p)L_c} + \frac{\omega_a}{f(\alpha_a) - (1-p)L_c} \right) > 0,$$

where $B > 0$ is the coefficient of $D\bar{x}_c$ in that substitution; the sign is an identity because moving both endpoints of the strip together changes every cell length by exactly $-L_c$. Hence

$$(f(\alpha_c) - L_c) D\alpha_c = -\ell(\underline{x}_c) + \ell(\bar{x}_c) D\bar{x}_c < -\ell(\underline{x}_c) + \ell(\bar{x}_c) = -L_c < 0 :$$

raising the floor never raises the adopter's own bar, and the second term of Φ'_c is positive. The first is positive whenever

$$D\alpha_b = \frac{(1-p) [g(\underline{x}_c) - g(\bar{x}_c) D\bar{x}_c]}{f(\alpha_b) - (1-p)L_c} < 1,$$

for which $(1-p) [g(\underline{x}_c) + g(\bar{x}_c) |D\bar{x}_c|] < f(\alpha_b) - (1-p)L_c$ suffices; at the baseline the left side never exceeds 0.27 and the right never falls below 0.92 on the admissible range. With two adopters, raising c 's floor removes spillover from c 's strip into b 's and a 's clearing conditions, so α_b and α_a rise, shrinking b 's gain cell and widening its loss cell: $\partial\Phi_b/\partial x_c < 0$. Raising b 's floor shortens b 's strip, so α_b and α_c fall, widening c 's gain cell and shrinking its loss cell: $\partial\Phi_c/\partial x_b > 0$.

(iii) *Uniqueness.* For the lone adopter, (i) makes Φ_c a well-defined function of the floor and (ii) makes it strictly increasing, so it has at most one zero. For two adopters, the Jacobian of (Φ_b, Φ_c) in the floors has positive diagonal by (ii) and off-diagonal entries of opposite sign, so its determinant is positive and it is a P-matrix on any rectangle of floors where the diagonals are positive; by the Gale–Nikaido theorem the floor system has at most one solution there. Existence is the computed fixed point, so the equilibrium for each set of adopters is unique.

The two frictions. With $s > 0$, Φ_c gains the term $-s(1-p)g$, whose derivative in the floor, $sp(1-D\alpha_b)$, has the sign of the first term of Φ'_c , and the cross-effects gain terms with the signs already established; with $q > 0$, Φ_c and every offer mass in Equation (19) are multiplied by $1-q$, which only relaxes the density condition. The argument is otherwise unchanged. $\square$

D.4 The taker–decliner achievement gap

In Proposition C.1, programs choose their unconditional-offer ranges knowing p and s . Even so, the students who take an offer differ from what their observables predict, and from those who decline it. Fix an adopter k and a target x in its range, and write $g_k(x)$ for the length of the gain cell, the share of offerees at x who would have placed above k .

Proposition D.6 (The taker–decliner achievement gap). In the equilibrium of [Proposition C.1](#), at every target x in adopter k 's unconditional-offer range:

- (i) Take-up is $(1 - q)(1 - pg_k(x))$: private information lowers it, moral hazard does not.
- (ii) Takers fall short of the ability their observables project by

$$\underbrace{\frac{pg_k(1 - g_k)}{2(1 - pg_k)}}_{\text{private information}} + \underbrace{s}_{\text{moral hazard}},$$

where the first term is positive if and only if $p > 0$. Either force alone produces the shortfall, and the same shortfall arises from a continuum of (p, s) pairs.

- (iii) Takers score below decliners by

$$\Delta_k(x) = \underbrace{\frac{pg_k(1 - g_k)}{2} \left[\frac{1}{1 - pg_k} + \frac{1 - q}{q + (1 - q)pg_k} \right]}_{\text{private information}} + \underbrace{s}_{\text{moral hazard}}.$$

Under private information the decliners are the informed students who would have placed above k ; under moral hazard alone they are the q who decline for other reasons, and the gap is s . Both forces produce the gap, and the gap does not apportion it.

- (iv) Adopters anticipate the shortfall and net it out in choosing the floor. Programs above the highest adopter do not choose, and their intakes fall below the benchmark; under the setup of [Figure C.1](#) the same is true of b when it adopts. Moral hazard is not redistributed: total attainment falls by s times the mass of takers.

At the calibration of [Figure C.1](#), $p = 0.104$ alone and $s = 0.057$ alone each produce a gap of -0.139SD , and together -0.278SD . Private information redistributes: at $p = 0.104$, c gains 0.040 in mean intake ability (0.050 with neither force), a loses 0.032, and b loses 0.025 although it adopts, because c 's offers capture the top of b 's band. Moral hazard destroys: at $s = 0.057$, c 's gain from adopting shrinks to 0.006, b withdraws and still loses 0.024, and total attainment falls by s times the mass of takers (the box in [Appendix D.3](#)).

D.5 Proofs of the propositions

Proof of Proposition C.1. Step 1: who adopts. a and d make no offers by the remark after [Equation \(18\)](#). For $k \in \{b, c\}$, [Equation \(18\)](#) gives $V_k \geq 0$ if and only if

$$(1 - p)g_k(x)^2 - \ell_k(x)^2 \geq 2s(1 - pg_k(x)),$$

which at $s = 0$ is $\sqrt{1 - p}g_k(x) \geq \ell_k(x)$, the gain cell at least $1/\sqrt{1 - p}$ times the loss cell. We take $s = 0$ for the closed forms and return to s in Step 2. Since V_k is strictly

increasing in x below the ceiling and nonpositive above it, the strictly profitable targets form the interval $(\underline{x}'_k(p), \bar{x}'_k)$ with the floor

$$\underline{x}'_k(p) = \frac{\alpha'_k - \sqrt{1-p} (1 - \bar{\alpha}'_k)}{1 + \sqrt{1-p}},$$

the break-even target, excluded by convention. At $p = 0$ the floor is $\underline{x}'_k = \frac{1}{2}(\alpha'_k + \bar{\alpha}'_k) - \frac{1}{2}$, the target whose two cells are equal; at the baseline, $\underline{x}'_c = 0.354 < \bar{x}'_c = 0.495$ and $\underline{x}'_b = 0.646 < \bar{x}'_b = 0.698$, so both middle programs adopt (C3).³⁸ The two ranges are disjoint, c 's below b 's: $\bar{x}'_c < \underline{x}'_b$ at the benchmark, and $\bar{x}'_c < \underline{x}'_b$ at every computed equilibrium (Appendix D.3). These are the unconditional-offer ranges of the marginal analysis; the equilibrium ranges differ, because they shift the thresholds (b 's is wider and c 's narrower at $p = s = 0$), and are reported in Appendix D.3.

Step 2: shrinking and exit. The floor is strictly increasing in s , and in p wherever $g_k > 2s$ at the floor, which holds throughout at the baseline (by Equation (18), V_k rises in x and falls in s and in p there), and the ceiling depends on neither, so the range shrinks from below as either force grows. At $s = 0$ it is nonempty exactly while the ceiling target still pays, $\sqrt{1-p} g_k(\bar{x}'_k) \geq \ell_k(\bar{x}'_k)$, i.e.

$$p \leq \hat{p}'_k = 1 - \left[\frac{\ell_k(\bar{x}'_k)}{g_k(\bar{x}'_k)} \right]^2.$$

At the baseline, $\hat{p}'_b = 0.443$ and $\hat{p}'_c = 0.816$, so the ladder is $\{b, c\} \rightarrow \{c\} \rightarrow \emptyset$. c 's exit is exact at any scale, $\hat{p}_c = \hat{p}'_c$, because no unconditional-offer range remains at c 's exit and the marginal offer faces the benchmark thresholds; b 's is not, since b exits into a market where c still makes unconditional offers, and Appendix D.3 computes $\hat{p}_b = 0.575$. In s , the ceiling target pays exactly while $s \leq \hat{s}'_k(p) = [(1-p)g_k^2 - \ell_k^2] / [2(1-p)g_k]$ evaluated at $\bar{x}'_k$, a frontier that is positive exactly for $p < \hat{p}'_k$ and decreasing in p wherever $g_k > 2s$; at $p = 0$ it is $G_k(\bar{x}'_k)$, 0.100 for c and 0.036 for b . As with p , c 's frontier is exact at scale and b 's is not: Appendix D.3 computes $\hat{s}_b = 0.060$ at $p = 0$. The order of exit is the same under both forces, and one number decides it. With equal capacities the bands have a common width $\bar{w} = \alpha'_a - \alpha'_b = \alpha'_b - \alpha'_c$, so at any target $\ell_k + g_k = 1 - \bar{w}$ for $k \in \{b, c\}$, and both exits are decreasing functions of the loss cell at the ceiling alone: $\hat{p}'_k = 1 - [\ell_k / (1 - \bar{w} - \ell_k)]^2$, and $\hat{s}'_k(p)$ has numerator $(1-p)(1 - \bar{w} - \ell_k)^2 - \ell_k^2$, falling in ℓ_k , over denominator $2[1 - p(1 - \bar{w} - \ell_k)]$, rising in it. Hence b exits first, in p and in s at every p , if and only if $\ell_b(\bar{x}'_b) > \ell_c(\bar{x}'_c)$: its ceiling lies deeper below its own threshold, because the insurance it sells, against falling to c , is worth less to an uninformed student than the insurance c sells, against falling to d . At the baseline $0.302 > 0.212$; the inequality, and with it both orders at once, reverses at $u_b = 2.63$. The nesting is a consequence

³⁸(C3) is a joint condition on preferences and geometry: a low enough u_k , or weak enough risk aversion, pushes $\bar{x}'_k$ below the floor and removes k from the adopter set. It fails for a under every preference, since $\bar{x}'_a > 1 \geq \bar{x}'_a$.

of equal capacities, not of the two forces. The s -intercept $G_k(\bar{x}'_k) = \frac{1}{2}(g_k^2 - \ell_k^2)$ scales with the size of the cells, while the p -intercept $1 - (\ell_k/g_k)^2$ depends only on their ratio; with a common $\ell_k + g_k$ the two move together. With unequal capacities $\ell_k + g_k$ differs across programs, a program can have both cells longer than the other's, and the frontiers cross: at capacities $(0.4, 0.2, 0.2, 0.2)$ and $u_b = 2.6$, c exits first in s ($\hat{s}_c(0) = 0.090 < \hat{s}_b(0) = 0.100$) but b exits first in p (the frontiers cross near $p = 0.6$), marginally and at scale alike; such crossings occur only within a narrow band of u_b around the point where the ordering flips.

Step 3: no offers beyond the exit, at any scale. At $p = 1$, consider any profile of unconditional-offer sets, of any size, and let k be the highest-ranked adopter. Every taker is informed, so k 's takers have $A < \bar{\alpha}_k$ (the remark after Lemma D.2), and the takers of lower adopters lie below their own upper thresholds, hence below $\bar{\alpha}_k$ as well. No taker has $A \geq \bar{\alpha}_k$, so the thresholds above k are the benchmark ones, and k 's intake is a set of mass $\frac{1}{4}$ inside $\{A < \bar{\alpha}_k\}$ that excludes the takers of other programs. Withdrawing its offers gives k the best such set, since conditional admission is assortative, so its payoff rises weakly, and strictly unless its takers already form the top of the set, in which case its offers have zero value and are excluded by convention. Hence no program adopts, and the outcome is Lemma D.1. If v_2 were public, an unconditional offer could condition on A and would be a conditional offer, so Lemma D.1 again applies. For $p \geq \max(\hat{p}_b, \hat{p}_c)$ at positive scale, or $s \geq \hat{s}_k(p)$ for each k , Lemma D.4 shows that any profitable unconditional-offer set is an interval, and the equilibrium system of Appendix D.3 has no positive-length solution there. The $p = 1$ argument holds a fortiori for $s > 0$, which only lowers the value of every taker.

Uniqueness. In the vanishing-size limit no unconditional-offer set moves the thresholds, so each program's payoff is additive across its targets and its unique best response is to offer to every strictly profitable target and no other; the conventions resolve the remaining indifferences (zero-value targets and the marginal student). The configuration of adopters is therefore unique at every p , and Lemma D.5 shows the same for the equilibrium levels at positive scale. $\square$

Proof of Proposition D.6. Fix a target x in k 's range, so $x < \bar{x}_k$, with cells evaluated at the equilibrium thresholds (the remark after Lemma D.2). Of the holders at x , a share q decline regardless; of the rest, the uninformed accept, and the informed accept exactly on the loss and neutral cells, $v_2 < 1 - g_k$, by Lemma D.2(ii). Take-up is therefore $(1 - q)[(1 - p) + p(1 - g_k)] = (1 - q)(1 - pg_k)$, which is (i); s does not enter because a taker's score is realized after she accepts.

For (ii), v_2 is uniform on $[0, 1]$ at every target, so the observable projection of a holder's v_2 is $\frac{1}{2}$. Takers have mass $(1 - q)(1 - pg_k)$ and mean

$$v_2 = \frac{(1 - p)\frac{1}{2} + p\frac{(1 - g_k)^2}{2}}{1 - pg_k} = \frac{1}{2} - \frac{pg_k(1 - g_k)}{2(1 - pg_k)},$$

and score s less than that. The private-information part is positive exactly when $p > 0$, since $g_k \in (0, 1)$ by (C2); the shortfall is continuous and increasing in both p and s , so each of its level sets in the (p, s) plane is a curve.

For (iii), decliners have mass $q + (1 - q)p g_k$: the q who decline regardless, with mean v_2 of $\frac{1}{2}$, and the informed on the gain cell, with mean $1 - g_k/2$; their mean

$$v_2 = \frac{\frac{q}{2} + (1 - q)p g_k \left(1 - \frac{g_k}{2}\right)}{q + (1 - q)p g_k} = \frac{1}{2} + \frac{(1 - q)p g_k (1 - g_k)}{2(q + (1 - q)p g_k)}.$$

Subtracting the takers' realized mean gives $\Delta_k(x)$. At $p = 0$ the decliners are the q alone and the gap is s ; at $q = 0$ the private-information term is $(1 - g_k)/[2(1 - p g_k)]$, which does not vanish as $p \rightarrow 0$, because the only decliners are then the informed on the gain cell however few they are. Δ_k is continuous and increasing in both p and s , so the same gap arises along a curve of (p, s) pairs.

The remaining claims are (iv). The floor condition $V_k(\underline{x}_k) = 0$ of [Lemma D.4](#) is stated in realized values, so the equilibrium range already nets out both the informed decliners and the s ; the shortfall is anticipated. Let j be a program above the highest adopter. Only the captured-from-above sum enters j 's condition in [Equation \(19\)](#), so $\underline{\alpha}_j < \underline{\alpha}'_j$ ([Appendix D.3](#)), and the same holds for every program above j , so $\bar{\alpha}_j \leq \bar{\alpha}'_j$. Its intake is therefore a set of mass $\frac{1}{4}$ inside $\{A < \bar{\alpha}'_j\}$ that excludes a positive mass of students with $A \geq \underline{\alpha}'_j$, the uninformed takers of the adopters below whose v_2 is high enough (at $x = 0.45$, $v_2 > 0.85$ reaches a 's benchmark band). The benchmark intake $[\underline{\alpha}'_j, \bar{\alpha}'_j]$ is the unique mass- $\frac{1}{4}$ subset of $\{A < \bar{\alpha}'_j\}$ with the highest mean, so j 's mean realized ability is below it. For b as an adopter the comparison is numerical: its mean intake ability is below the benchmark in panels (a) and (b) of [Figure C.1](#) and at every p in the table of [Appendix D.3](#). Finally, every student enrolls somewhere and capacities sum to one, so the sum of realized intake abilities is the total ability of the population less s times the mass of takers. $\square$

E Optimality of the Application Portfolio under Correlated Admissions

The demand-side estimator of Section 4.3 treats the observed portfolio as satisfying the one-shot-swap (OSS) inequalities of Larroucau and Rios (2020). As noted in Section 4.3.2, these inequalities are *necessary* for optimality but not, in general, *sufficient*: under independent admissions they exactly characterize the optimum of the portfolio problem, because the Marginal Improvement Algorithm (MIA) of Chade and Smith (2006) solves it, but under the correlated admissions of Assumption 2 the MIA is only an approximation, and the exact optimality conditions are the dynamic-programming inequalities of Ali and Shorrer (2025). This appendix makes the relationship between the two precise. We characterize the gap between the OSS inequalities and the exact conditions, show that it vanishes under independence, and give a closed-form bound under the common-score structure of Section 4.1. The bound is small at the residual offer correlation we estimate, which justifies our use of the OSS inequalities in estimation.

E.1 Setup

Fix a student and drop the subscript i . Following Implication 1, the utility from program s is its Stage-1 utility $u_s \equiv U_{is}$, and the probability attached to it is $q_s \equiv q_{is}$, the joint probability that s makes her an offer and admits her (Section 4.3.1). As in Implication 1, admission events are comonotone: a student admitted to a program is also admitted to every program with a higher q_s .

Comonotone events with these marginals are generated by a single score. Let S be uniform on $[0, 1]$ with CDF F , and let the student be admitted to s if and only if $S < \tau_s$, where the cutoff is $\tau_s \equiv q_s$. Then $F(\tau_s) = q_s$, and a more selective program has a lower cutoff.

Because one score governs every admission, rejection is informative. Define the posterior belief after rejection by program s' as

$$F_{s'} \equiv F(\cdot \mid S \geq \tau_{s'}), \quad (20)$$

so that for any program s less selective than s' ($\tau_s > \tau_{s'}$), the admission probability given rejection by s' is

$$F_{s'}(\tau_s) = \frac{F(\tau_s) - F(\tau_{s'})}{1 - F(\tau_{s'})}.$$

After rejection by every program in a set $\mathcal{R}$, the posterior is $F(\cdot \mid S \geq \max_{s \in \mathcal{R}} \tau_s)$, which depends only on the most selective program that rejected the student.

For a portfolio $A = \{s_1, \dots, s_m\}$ with $\tau_{s_1} \leq \dots \leq \tau_{s_m}$ and belief G , the expected value of the best admission is

$$V(A; G) = \int \max_{\substack{s \in A \\ S < \tau_s}} u_s dG(S) = \sum_{l=1}^m [G(\tau_{s_l}) - G(\tau_{s_{l-1}})] \max_{j \geq l} u_{s_j}, \quad (21)$$

with $G(\tau_{s_0}) \equiv 0$. At $G = F$ this is the portfolio value of Implication 1. Let $A^* = \{s_1^*, \dots, s_K^*\}$ be the observed portfolio, indexed so that $\tau_{s_1^*} \leq \dots \leq \tau_{s_K^*}$; by Implication 2, $u_{s_1^*} \geq \dots \geq u_{s_K^*}$, so the inner maximum in Equation (21) is $u_{s_l^*}$. Let $\tilde{V}(m, G) = \max_{|A|=m} V(A; G)$ be the value of the best m -program portfolio under belief G , and let $A_{m+1:K}^* = \{s_{m+1}^*, \dots, s_K^*\}$ be the observed portfolio below level m .

E.2 The two sets of inequalities

E.2.1 Larroucau–Rios, independent admissions (LR)

When admissions are independent across programs, the MIA of [Chade and Smith \(2006\)](#) solves the portfolio problem, and [Larroucau and Rios \(2020\)](#) show that optimality of A^* can be characterized by a polynomial-sized set of one-shot-swap inequalities: for each $s \in A^*$ and each $\ell \notin A^*$,

$$V(A^*; F) \geq V((A^* \setminus \{s\}) \cup \{\ell\}; F). \quad (\text{LR})$$

These $K(|\mathcal{S}| - K)$ inequalities compare portfolios that differ in a single element, each evaluated at the *same* prior F . Because $F(\tau_s) = q_s$, they are the inequalities of [Equation \(16\)](#).

E.2.2 Ali–Shorrer, correlated admissions (AS)

When admissions are correlated through the common score, the optimal portfolio instead satisfies the dynamic-programming recursion of [Ali and Shorrer \(2025\)](#). Optimality of A^* requires that, at each level $m \in \{1, \dots, K\}$, the choice s_m^* be optimal conditional on rejection by all more selective programs already chosen. Let $\tau_{s_0^*} = 0$ so that $F_{s_0^*} = F$. Then, for each level m and each eligible alternative j with $\tau_j > \tau_{s_{m-1}^*}$,

$$\begin{aligned} F_{s_{m-1}^*}(\tau_{s_m^*}) u_{s_m^*} + [1 - F_{s_{m-1}^*}(\tau_{s_m^*})] \tilde{V}(K - m, F_{s_m^*}) \\ \geq F_{s_{m-1}^*}(\tau_j) u_j + [1 - F_{s_{m-1}^*}(\tau_j)] \tilde{V}(K - m, F_j). \end{aligned} \quad (\text{AS})$$

The left-hand side reflects that, under the posterior $F_{s_{m-1}^*}$, program s_m^* admits the student with probability $F_{s_{m-1}^*}(\tau_{s_m^*})$, and upon rejection the student continues *optimally* under the updated posterior $F_{s_m^*}$. The right-hand side is the analogous value from choosing j in place of s_m^* at level m .

E.3 The gap

The LR inequalities hold the observed tail $A_{m+1:K}^*$ fixed; the AS inequalities reoptimize the continuation after rejection. To compare them, condition on rejection by the programs $s_1^*, \dots, s_{m-1}^*$, which both portfolios share, and write the swap of s_m^* for j with the tail held fixed:

$$\begin{aligned} F_{s_{m-1}^*}(\tau_{s_m^*}) u_{s_m^*} + [1 - F_{s_{m-1}^*}(\tau_{s_m^*})] V(A_{m+1:K}^*; F_{s_m^*}) \\ \geq F_{s_{m-1}^*}(\tau_j) u_j + [1 - F_{s_{m-1}^*}(\tau_j)] V(A_{m+1:K}^*; F_j). \end{aligned} \quad (\text{LR}')$$

When $\tau_{s_{m-1}^*} < \tau_j \leq \tau_{s_{m+1}^*}$ and $u_j \geq u_{s_{m+1}^*}$, (LR') is the one-shot swap of s_m^* for j in [Equation \(LR\)](#), divided by $1 - \tau_{s_{m-1}^*}$. Comparing [Equation \(LR'\)](#) to [Equation \(AS\)](#), the right-hand side of (AS) exceeds that of (LR') by

$$\Delta_m(j) \equiv [1 - F_{s_{m-1}^*}(\tau_j)] [\tilde{V}(K - m, F_j) - V(A_{m+1:K}^*; F_j)] \geq 0, \quad (22)$$

since $\tilde{V}(K - m, F_j)$ is the value of the *optimal* $(K - m)$ -program portfolio under F_j , which weakly exceeds the value of the fixed tail $A_{m+1:K}^*$ under the same belief. The left-hand sides coincide when A^* satisfies (AS) at every level: by backward induction the observed tail is then the optimal continuation, $\tilde{V}(K - m, F_{s_m^*}) = V(A_{m+1:K}^*; F_{s_m^*})$. Hence (AS) implies (LR'), and (LR') is slack by exactly

$\Delta_m(j)$: the OSS inequalities check only whether a single swap improves on the observed portfolio holding the rest fixed, whereas the exact condition allows the continuation to be reoptimized after the swap.

We describe two properties of $\Delta_m(j)$.

1. *Zero under independence.* When admissions are independent, rejection is uninformative and the OSS inequalities characterize the optimum exactly (Larroucau and Rios, 2020); the gap arises only because rejection is informative.
2. *Scaled by the rejection probability.* $\Delta_m(j)$ is scaled by $1 - F_{s_{m-1}^*}(\tau_j)$, the probability that program j rejects the student under the level- $(m-1)$ posterior. Because the bracket in Equation (22) is at most $\max_s u_s$, $\Delta_m(j) \leq [1 - F_{s_{m-1}^*}(\tau_j)] \max_s u_s$. For safety programs with high admission probability, this factor is small regardless of the reoptimization gain.

E.4 Closed form under the common-score structure

Under the common-score structure of Section 4.1, the gap admits a closed form. Because S is uniform, $F(\tau) = \tau$ for $\tau \in [0, 1]$. The posterior after rejection by s' is then the truncated uniform

$$F_{s'} = U[\tau_{s'}, 1], \quad F_{s'}(\tau_s) = \max \left\{ \frac{\tau_s - \tau_{s'}}{1 - \tau_{s'}}, 0 \right\}.$$

For $\tau_j \leq \tau_{s_{m+1}^*}$, the expected utility of the tail portfolio $A_{m+1:K}^*$ under $F_j = U[\tau_j, 1]$ is

$$V(A_{m+1:K}^*; F_j) = \frac{1}{1 - \tau_j} \left[u_{s_{m+1}^*}(\tau_{s_{m+1}^*} - \tau_j) + \sum_{l=m+2}^K u_{s_l^*}(\tau_{s_l^*} - \tau_{s_{l-1}^*}) \right],$$

(for a less selective j , the tail programs more selective than j drop out), so that the gap Equation (22) becomes

$$\Delta_m(j) = \frac{1 - \tau_j}{1 - \tau_{s_{m-1}^*}} \left[\tilde{V}(K - m, U[\tau_j, 1]) - V(A_{m+1:K}^*; U[\tau_j, 1]) \right]. \quad (23)$$

The magnitude of Equation (23) is governed by how dispersed the cutoffs $\{\tau_s\}$ are, which in turn is governed by the importance of the common component μ_i relative to the student's baseline characteristics. When μ_i carries little weight, the cutoffs cluster near 0 and 1: near 1, posterior updates upon rejection are small, so the bracketed reoptimization gain is small; near 0, the rejection probability $1 - \tau_j$ multiplying the gain is small. Either way $\Delta_m(j)$ is small. When the common component is more important, the cutoffs spread out and the gap widens.

In our setting the common component is empirically minor. The average residual pairwise offer correlation is 0.06, rising to roughly 0.12 among top-tercile universities, so the cutoffs are only weakly dispersed and the OSS inequalities are a close approximation to the exact Ali–Shorrer conditions. Equivalently, one may interpret the estimator as identifying preferences from the requirement that the observed portfolio be a *local* optimum of the portfolio problem — for which the OSS inequalities are necessary — rather than a global one, with Equation (23) bounding the difference between the two.

F Structural Model Estimation Details

This appendix records how the demand-side estimator of Section 4.3 is implemented: the construction of the revealed-preference cone, the Gibbs sampler, the treatment of estimated admission beliefs, and the warm start. We write $\mathcal{L}_i(p)$ for the matrix of student i 's revealed-preference inequalities (the constraints $\mathcal{L}_i$ of Section 4.3.2) evaluated at the offer probabilities p , so the augmented utilities satisfy the cone $\mathcal{L}_i(p) U_i > 0$.

F.1 Constraint construction

For each student the observed portfolio implies the cone of Section 4.3.2: the ranking rows (utilities ordered inversely to the offer probabilities $\hat{p}_{is}$, by Implication 2), the positivity rows, and the one-shot-swap rows of Equation (16), using the inequality reduction whereby each of the (at most five) chosen slots may be swapped for any non-chosen program, so the comparison set is the complement of the portfolio rather than the set of full portfolios. Each swap row constrains the chosen utilities together with exactly one swapped-in alternative (a single negative coefficient), so we store the chosen block densely and the alternatives as a per-row index of that one nonzero entry, letting the utility draw touch each alternative once per sweep. The swap rows use the total acceptance probabilities $\hat{q}_{is} = \hat{p}_{is}[\hat{g}_{is} + (1 - \hat{g}_{is}) \hat{a}_{is}^o]$, the joint probability of Section 4.3.1, from the Firth-penalized offer and conversion probits of Section 4.3.1: both the offer probability and the conversion probability enter the portfolio stage, consistent with Implication 1. For the ranking stage we additionally form, for each observed pair of ranked choices, the probability that the second choice converts given the first does not, by truncating the common μ_i above the first choice's threshold in Equation (8).

F.2 Gibbs sampler

We draw from the joint posterior of $(\delta, \bar{\delta}, \sigma_\delta^2, \bar{\beta}^c, \Gamma^c, \sigma_\eta^2, \sigma_v^2, \{\eta_i^c\}, \{U_i\})$ by a blocked sampler. Each sweep alternates:

- (a) **Utilities** $U_i \mid \delta, \beta_i^c, \sigma_v^2$, truncated to $\mathcal{L}_i(\hat{p}) U_i > 0$: conditional on the chosen utilities the alternatives are independent and drawn as exact one-dimensional truncated normals, while the (at most five) chosen utilities are drawn jointly by hit-and-run within the cone. The program fixed effects δ and the coefficient $\beta_i^c = \bar{\beta}^c + \Gamma^c \text{Low-SES}_i + \eta_i^c$ enter only through this draw's mean.
- (b) **Program fixed effects** $\delta \mid U, \beta^c, \sigma_v^2$: a conjugate Normal-means update. The residual $U_{is} + d_{is} - \beta_i^c$ close_{is} is a noisy observation of δ_s with precision σ_v^{-2} , so δ_s is drawn from a Normal with posterior precision $n_s \sigma_v^{-2} + \sigma_\delta^{-2}$, where n_s counts the latent utilities in which s appears. The hyperparameters of the prior $\delta_s \sim N(\bar{\delta}, \sigma_\delta^2)$ are themselves drawn (Normal and inverse-Gamma), shrinking thinly-applied-to programs toward the population mean rather than letting them take extreme values fit to a handful of students.
- (c) **Preference parameters**: η_i^c (Normal), $(\bar{\beta}^c, \Gamma^c)$ in one joint Normal draw, σ_v^2 (inverse-Gamma), and σ_η^2 under the half- t prior described below. The nesting of Equation (13) is imposed by

masking: η^c and σ_η^2 are sampled only in the specification carrying an i.i.d. component, and Γ^c only where the close coefficient loads on Low-SES status.

The cone is homogeneous in U , so the scale is pinned by fixing the linear distance coefficient at $\beta^d = -0.01$ and estimating σ_v^2 freely, which denominates utilities in units of 100 kilometers of travel. For the heterogeneity scales we use a weakly informative half- t prior and update the $(\eta_i^c, \sigma_\eta^2)$ block by an ancillarity–sufficiency interweave — one centered and one non-centered update per sweep — which prevents the funnel between η_i^c and σ_η^2 from inflating or collapsing the variance. The same interweave is applied to the $(\delta, \sigma_\delta^2)$ block: the centered parameterization mixes well for heavily-applied-to programs and the non-centered one for thin programs, so interweaving keeps neither end of the program-size distribution in control of the chain.

F.3 Estimated beliefs

On real data $\mathcal{L}_i(\hat{q})$ is built from the estimated joint acceptance probabilities $\hat{q}_{is}$ of [Section 4.3.1](#), which enter the row *coefficients*, so sampling error that reorders two nearby programs mis-specifies rows. As in [Section 4.3.2](#) we treat q as latent and resample it jointly with U , keeping every utility draw over an exact cone. We augment the sampler with an independence-Metropolis step: propose q' , recomputed from the same joint formula, by drawing the offer-probit parameters (γ_s, c_s) from their asymptotic distribution, holding the conversion-probit parameters $(\gamma_s^{\text{admit}}, \alpha_s, c_s^R)$ fixed at their point estimate, then accept iff the observed portfolio is still optimal under the proposal — with this proposal the Metropolis ratio reduces to that feasibility indicator — rebuilding the rows for accepted students and otherwise keeping the incumbent. The test is reorder-aware, and initializing at $\hat{q}$ nests the fixed-belief sampler.

F.4 Warm start

The free-scale sampler is initialized from a pooled preliminary run in which all students share one close-to-home coefficient — a fast-mixing special case of the same machinery — which supplies a population mean near the truth, per-student deviations from a ridge of each student’s pooled utilities on their characteristics, and a feasible utility configuration so the first constrained draw does not throw the excursions that seed a divergence. The fixed effects are started at the program-level means of the pooled utilities, shrunk toward the grand mean by a factor decreasing in n_s , so that thinly-applied-to programs start near $\bar{\delta}$ and heavily-applied-to ones start near their own mean; this is already close to the posterior mode and spares the δ block a long transient. The pooled run is estimated on a free scale and handed off by a single global rescaling onto $\beta^d = -0.01$, exact by scale-invariance of the cone.

F.5 Ranking stage

The ranking stage is estimated by the same device: each candidate ranking’s expected value is linear in the program latents, so “the realized pair is the expected-utility maximizer” gives linear winner inequalities $\mathcal{W}_i U_i \geq 0$ sampled with the same hit-and-run kernel. Conditional on the latents we draw the ranking-shock variance σ_ϵ^2 (inverse-Gamma) and the coefficient λ_i on the additively-entering unconditional-offer premium (a multivariate-normal regression on z_i of the utility gap that

carries the unconditional-offer premium, as in [Equation \(9\)](#)). This stage also carries the private signal: ψ_i is augmented per student and σ_ψ^2 drawn from its conjugate posterior given $\{\psi_i\}$ and the realized $\{r_i\}$. Because ψ enters the ranking inequalities through $\Phi(\cdot)$ rather than linearly, the ψ_i draw is a Metropolis step within the block rather than a truncated normal. Stage-1 uncertainty is propagated as a cut: we resample each β_i^c , and the fixed effects δ , from their Stage-1 posterior but never let the ranking data revise them. Nothing in this stage changes under the fixed-effects specification, since it conditions on the Stage-1 utilities rather than on how they were built; unconditional-offer status varies within program, so λ_i is not collinear with δ .

We validate on data simulated to match the model — first against the simulator’s oracle probabilities to isolate the sampler, then with the estimated probits and the resampling step of [Appendix F.3](#). The close-to-home coefficients and the heterogeneity parameters recover in both; the common level of the fixed effects, identified only through the participation margin ([Section 4.4](#)), is the most demanding. [Appendix F.6](#) gives the convergence-diagnostic methodology.

F.6 Validity of the sequential estimator

Within each block the estimator is a data-augmentation Gibbs sampler of the kind [McCulloch and Rossi \(1994\)](#) introduced for discrete choice, with Metropolis steps inside the block where the full conditional is not conjugate: the belief resampling at Stage 1 and the signal draw at Stage 3. The resulting Metropolis-within-Gibbs kernel leaves the target posterior invariant and is ergodic under the conditions of [Tierney \(1994\)](#). We report posterior means and standard deviations after burn-in and assess convergence by Gelman–Rubin $\hat{R}$, effective sample sizes and Geweke diagnostics.

Across blocks, each conditions only on objects recovered earlier, which is a deliberate cut rather than joint estimation. It rests on beliefs being pinned at their true values ([Section 4.1.2](#)) together with the fact that Stage-1 beliefs depend on the supply side only through the scalar q_{is} , whose two components recover correctly irrespective of the correlation structure because the common component is integrated out of the relevant marginal. The demand-side posteriors treat supply-side probabilities as known data, with no feedback from preferences to admissions.

The sequence is one of estimation order rather than stage order: the conversion probit is fit on first-ranked choices, a Stage-3 decision, and its fitted values enter Stage-1 beliefs. There is no circularity, since $\hat{q}_{is}$ is evaluated for every applicant rather than at the student’s own realized ranking, but the estimator is not recursive in the game’s timing, and the price is that the conversion probit is fit only on students who rank the program (the “at-risk” sample of [Section 4.3.1](#)).

F.7 Run lengths and priors

The Stage-1 block is a Gibbs sampler over $(\delta, \bar{\delta}, \sigma_\delta^2, \bar{\beta}^c, \Gamma^c, \sigma_\eta^2, \sigma_v^2, \{\eta_i^c\})$ and the utilities $\{U_i\}$, run for 50,000 iterations with 5,000 discarded as burn-in. Priors are Normal-inverse-Gamma for $(\delta, \sigma_\delta^2)$, Normal for $(\bar{\beta}^c, \Gamma^c)$ with a half- t prior on σ_η , and inverse-Gamma for σ_v^2 . The Stage-3 block is run for 250,000 iterations with 25,000 discarded.

F.8 Counterfactuals

We detail the application system designs utilized in [Section 4.6](#).

For each program-cohort we compute the *expected capacity* $\hat{K}_{st}$ that program s expects to fill in cohort t , given the offers it made at Stage 2 in the estimated baseline and its own beliefs about yield and cutoff clearing. Summing over the applicants who hold an offer from s , this is the probability that an applicant ranks s first and clears its condition, plus the probability that she ranks s second, misses her first-choice condition, and clears s 's. The second term is computed as a joint rather than as a product of marginals: missing one condition and clearing another are both driven by the same realization of r_i , so the two events are negatively dependent and multiplying the marginals overstates the rebound. We hold $\hat{K}_{st}$ fixed across designs.

For each design, we compute a fixed point of the best-response map in student and program strategies. We iterate until three conditions hold: the weighted average gap between each program's expected intake and $\hat{K}_{st}$ is below a tolerance, aggregate intake is within tolerance of total capacity, and every program's intake is within three sampling standard deviations of $\hat{K}_{st}$. We cap the loop at 80 iterations and keep the best iterate among those that meet these conditions. Students choose portfolios at Stage 1 and rankings at Stage 3 to maximize expected utility, exactly as in [Section 4.1.4](#) and [Section 4.1.2](#), taking the cutoffs and the admission probabilities they imply as given. Programs choose scalar cutoffs, taking the applicant pool as given, to maximize the expected sum of enrolling students' quality subject to expected enrollment not exceeding $\hat{K}_{st}$, where a student's quality to program s is measured by s 's own estimated weights on her characteristics.

Two details of the program side matter. First, the relevant index differs by stage, because the program's information does. At Stage 2 the program has not yet observed r_i — that is what a conditional offer is for — so it maximizes expected quality conditional on its Stage-2 information, and the offer cutoff $\bar{\pi}_s$ is a threshold on π_{is} of [Equation \(11\)](#). At Stage 4, r_i is realized, and the conversion cutoff $\bar{\pi}_s^R$ is a threshold on the admissions index of [Equation \(8\)](#). Second, because the program's choice variable is a scalar threshold on a given index, maximizing expected enrolled quality subject to capacity is equivalent to using a threshold rule, following the argument of [Kapor \(2025\)](#).

Except where noted, the data-generating process in each design is identical to the setup described in [Section 4](#). Program fixed effects $\{\delta_s\}$ are recalibrated in the simulated baseline to match observed application counts and then held fixed across designs; all other preference parameters are held at their estimated values, and the choice set consists of the same set of programs as used in parameter estimation.

In simulation, the offer score of [Equation \(11\)](#) is implemented as $\pi_{is} = \kappa_s \tilde{z}_{is} + \mu_i$, where $\tilde{z}_{is}$ is the index $z_i^{\text{offer}} \gamma_s$ centered within each program-cohort and κ_s is a per-program scale fit to how sharply offers separate offer holders from other applicants in the data. The offer threshold varies by program-cohort, $\bar{\pi}_{st}$, and the offer thresholds L_{is} that students update on at Stage 3 are defined accordingly.

1. **Ban on unconditional offers.** In Stage 2 we set $\bar{\pi}_s^U = \infty$ for all program-cohorts, so that every offer is conditional and every first choice is tested against $\bar{\pi}_s^R$. Programs re-optimize $\bar{\pi}_s$ and $\bar{\pi}_s^R$.
2. **Unconditional behavior consistent with 2018.** In Stage 2, $\bar{\pi}_s^U$ is set program-cohort by program-cohort so that the share of s 's offers that are unconditional equals its 2018 share; what is held fixed across cohorts is the share rather than the threshold, since the applicant

pool differs by cohort. Programs re-optimize $\bar{\pi}_s$ and $\bar{\pi}_s^R$ subject to that constraint.

3. **Only unconditional offers.** In Stage 2 we set $\bar{\pi}_s^U = \bar{\pi}_s$ for all program-cohorts, so that every offer a program makes is unconditional and no admission is contingent on realized age-18 performance. $\bar{\pi}_s^R$ is then never applied, and programs re-optimize $\bar{\pi}_s$ alone. Admission is still rationed: the design removes the grade condition, not the offer threshold.
4. **Post-qualification admissions.** There is a single application stage, taken after age-18 results are released, so that r_i is common knowledge. Three things follow. The signal ψ_i is moot, since students and programs observe r_i alike. The conditional–unconditional distinction disappears, since there is no future performance left to condition on: every place a program offers is unconditional, and the single cutoff $\bar{\pi}_s^R$ on the admissions index of Equation (8) is the admission decision. Because application and ranking now coincide, a student knows her full idiosyncratic taste $v_{is} + \varepsilon_{is}$ when she chooses, rather than only v_{is} . Programs rank the full pool of applicants by that index and admit to capacity; students rank programs; the outcome is the stable matching, which we compute as the student-proposing deferred acceptance outcome. The only uncertainty a student faces at the point of choosing is her own μ_i , which programs observe and she does not. We evaluate the design both with student portfolios capped at five and with the cap relaxed, isolating the role of the portfolio constraint in the status-quo design.

Design 4 is also the policy analogue of removing students' informational advantage: once results are public, the asymmetry that drives selection into unconditional offers disappears institutionally rather than by assumption.